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An insurance risk model with stochastic volatility

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ABSTRACT

In this paper, we extend the Cramér–Lundberg insurance risk model perturbed by diffusion to incorporate stochastic volatility and study the resulting Gerber–Shiu expected discounted penalty (EDP) function. Under the assumption that volatility is driven by an underlying Ornstein–Uhlenbeck (OU) process, we derive the integro-differential equation which the EDP function satisfies. Not surprisingly, no closed-form solution exists; however, assuming the driving OU process is fast mean-reverting, we apply the singular perturbation theory to obtain an asymptotic expansion of the solution. Two integro-differential equations for the first two terms in this expansion are obtained and explicitly solved. When the claim size distribution is of phase-type, the asymptotic results simplify even further and we succeed in estimating the error of the approximation. Hyper-exponential and mixed-Erlang distributed claims are considered in some detail.

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1. Introduction

The Cramér–Lundberg risk model has been studied extensively in ruin theory—see e.g. Gerber and Shiu (1998), Klugman et al. (2004), Lin and Willmot (1999), Lin and Willmot (2000) and references therein. Dufresne and Gerber (1991) extend the Cramér-Lundberg risk model by adding a diffusion process. The additional diffusion term may be interpreted as the future uncertainty of aggregate claims, the future uncertainty of premium incomes, or the fluctuation of investment of surplus. This new model is usually called the perturbed compound Poisson risk model (with constant volatility). Dufresne and Gerber (1991) mainly focus on the probability of ruin caused by oscillation or by a claim. Cai and Yang (2005) study ruin with both constant and stochastic interest force, and demonstrate the ruin probability is always twice continuously differentiable provided the claim size has a continuous density function.

Traditionally, authors analyze the ruin probability and the distribution of deficit at ruin separately. Gerber and Landry (1998) generalize the discussion of Dufresne and Gerber (1991) to study the expected discounted deficit at ruin. They show that it satisfies a defective renewal equation and provide a natural probabilistic interpretation of this equation. Gerber and Shiu (1998) further introduce an expected discounted penalty function (henceforth referred to as the EDP function) which depends on the time of ruin, the surplus immediately before ruin and the deficit at ruin to analyze these and related quantities in a unified manner. Moreover, they demonstrate that under the classical compound Poisson model, this new EDP function also satisfies a defective renewal equation. The EDP function has proven to be a powerful tool for analyzing ruin related quantities from a large class of risk models. Tsai and Willmot (2002) study the EDP function for the perturbed compound Poisson risk model. When the claim size is of phase-type, Ren (2005) applies and extends the results of Lin and Willmot (1999) and Lin and Willmot (2000) to obtain matrix-type representations of the expected value of the time of ruin and the moments of the discounted deficit at ruin.

Chen et al. (2007) take an alternative approach to the EDP function and instead derive an integro-differential equation (IDE) which it satisfies. When the claim size distribution is phase-type, they use an ODE approach to solve the IDE and obtain a simple explicit expression. As an application, they work out an example in Leland's (1994) structural model of corporate default with downward jumps. However, Chen's approach requires the difficult task of finding all roots of the Lundberg fundamental equation. Other attempts at generalizing the perturbed compound Poisson risk model can be found in Yang and Zhang (2001) (for ruin probabilities) and Garrido and Morales (2006) (for the EDP function). In these works, the authors use spectrally negative Lévy processes to model heavy tailed behavior of the investment return.

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In this paper, we consider a perturbed compound Poisson risk process with stochastic volatility. The model may be viewed as an extension of the model in Chen et al. (2007). However, unlike the classical model or the spectrally negative Lévy extensions, our model allows for both upward and downward heavy tailed movements of the investment return through stochastic volatility.

The classical Black and Scholes (1973) model is well known to fail at explaining either historical price behavior or implied volatilities from options data. To correct some of these deficiencies, Heston (1993) assumes that volatility is driven by a Feller process (also known as the CIR process) to introduce stochasticity in volatility itself. An alternative approach is taken by Fouque et al. (2000), who instead model volatility as function of an underlying Ornstein–Uhlenbeck (OU) process. This second approach has the advantage of highly efficient calibration; in addition, it provides explicit closed form *approximate* pricing results—the approximation is exact in a particular homogenization limit to be discussed later on. In this paper, we extend the perturbed Cramér–Lundberg risk model in the spirit of Fouque et al. (2000) by assuming that volatility is driven by an underlying OU process. We then analyze the EDP function under this stochastic volatility (SV) extended risk model and provide explicit closed form results.

Through martingale methods, we obtain an IDE which the EDP function satisfies. This equation is the analog of the Chen et al. (2007) result for the perturbed compound Poisson risk model; however, it is much richer in structure, as it accounts for stochastic volatility, and in particular is highly non-linear. As a consequence, no explicit closed-form solution exists; nonetheless, through singular perturbation theory we are successful in computing the first few terms in an asymptotic expansion of the EDP function, where the inverse rate of mean-reversion acts as a small parameter.

In the financial literature, singular perturbation theory has been used to price and hedge derivatives with stochastic volatility in several contexts: Fouque et al. (2000) use it to obtain asymptotic pricing and hedging of European and American options; Cotton et al. (2004) apply the techniques to value interest rate derivatives; while, Hikspoors and Jaimungal (2008) utilize the method for valuing commodity derivatives. In this paper, by assuming that the volatility in the perturbed compound Poisson model is a fast mean-reverting process, singular perturbation theory is applied to approximate the EDP function. We derive two related IDEs for the first two terms of the asymptotic expansions and solve these equations in closed-form. When the claim size distribution is phase-type, the explicit expression of the asymptotic result is presented in matrix-type form and the approximation error bound is found.

The remainder of this paper is organized as follows. In Section 2, we review some known results for the EDP function in the perturbed compound Poisson risk model, as well as some properties of phase-type distributions. The results of Ren (2005) are extended here to provide a matrix-type expression for the EDP function for phase-type distributed claim sizes. In Section 3, we formulate our new stochastic volatility extended risk model, review the properties of OU processes and provide some background on singular perturbation theory. Section 4 derives the integro-differential equation for the EDP function in our new risk model. In Section 5, singular perturbation theory is applied to obtain an asymptotic expansion of the solution for this non-linear IDE. We especially focus on the results for phase-type distributed claim sizes. When the claim size distribution is hyper-exponential, we extract all roots of the Lundberg fundamental equation and provide the explicit form of the EDP function approximation. When the claim distribution is mixed-Erlang, to avoid the computation of a matrix exponential, we employ the results in Willmot and Woo (2007) and express the approximated terms as mixed-Erlang type. Finally, in Section 6 we provide some concluding remarks.

2. Some known results on jump-diffusion risk process

To introduce the perturbed compound Poisson risk model formally, let $\{Y_i; i = 1, 2, \ldots\}$ be independent and identically distributed (i.i.d.) positive random variables representing the successive individual claim amounts. These random variables are assumed to have common cumulative distribution function (c.d.f.) $P(y) = 1 - \overline{P}(y)$, $y \geq 0$ with continuous probability density function (p.d.f.) $p(y) = P'(y)$ and Laplace transform $\widetilde{p}(s) = \int_0^\infty \mathrm{e}^{-sy} p(y)\mathrm{d}y$. Furthermore, let the total number of claims up to time t, denoted by $N(t)$ and independent of $\{Y_i; i = 1, 2, \ldots\}$, be a Poisson process with parameter $\lambda > 0$. The aggregate claims process is thus $\{S_t = \sum_{i=1}^{N(t)} Y_i; t \geq 0\}$ where $S_t = 0$ if $N(t) = 0$.

Let U_t be the insurer's surplus at time $t \geq 0$, where $U_0 = u$ is initial surplus, and let the premium rate be $c = (1+\theta)\lambda\mathbb{E}Y_1$, where $\theta > 0$ is the relative safety loading. The surplus process $\{U_t; t \geq 0\}$ of the perturbed compound Poisson risk process is given by

$$U_t = u + ct - S_t + \sigma W_t, \quad t \geq 0, \tag{2.1}$$

where volatility $\sigma > 0$ and $\{W_t; t \geq 0\}$, independent of $\{S_t; t \geq 0\}$, is a one-dimensional standard Brownian motion.

As usual, the time of (ultimate) ruin is defined as $\widetilde{\tau} = \inf\{t \geq 0 : U_t \leq 0\}$, where $\widetilde{\tau} = \infty$ if ruin does not occur in finite time. According to the definition of ruin time, ruin is caused either by oscillation (diffusion) or by a claim.

Gerber and Landry (1998) study the EDP function

$$\begin{aligned}\Psi(u) &= \mathbb{E}[\mathrm{e}^{-\delta\widetilde{\tau}}\omega(U(\widetilde{\tau}))\,\mathbb{I}(\widetilde{\tau} < \infty)|U_0 = u] \\ &= \omega(0)\mathbb{E}[\mathrm{e}^{-\delta\widetilde{\tau}}\,\mathbb{I}(\widetilde{\tau} < \infty, U(\widetilde{\tau}) = 0)|U_0 = u] \\ &\quad + \mathbb{E}[\mathrm{e}^{-\delta\widetilde{\tau}}\omega(U(\widetilde{\tau}))\,\mathbb{I}(\widetilde{\tau} < \infty, U(\widetilde{\tau}) < 0)|U_0 = u],\end{aligned}$$

where $\delta > 0$ is interpreted as the force of interest or the variable of a Laplace transform, ω is a non-negative function defined on $(-\infty, 0]$, and $\mathbb{I}(\mathscr{D})$ is the indicator function of an event $\mathscr{D}$. It is important to point out that the penalty function ω is not necessarily continuous at 0.

Chen et al. (2007) derive the following IDE for $\Psi(u)$:

$$\begin{cases}(\mathcal{L}_c(\sigma) - \delta)\Psi(u) = 0, & u > 0, \\ \Psi(u) = \omega(u), & u \leq 0,\end{cases} \tag{2.2}$$

where the operator $\mathcal{L}_c$ is the infinitesimal generator of the surplus process $\{U_t; t \geq 0\}$ defined by

$$\begin{aligned}\mathcal{L}_c(\sigma)h(u) &\triangleq \frac{\sigma^2}{2}h''(u) + ch'(u) \\ &\quad + \lambda\int_0^\infty h(u-z)p(z)\mathrm{d}z - \lambda h(u),\end{aligned} \tag{2.3}$$

for arbitrary functions $h \in C^2(\mathbb{R})$.

Denote $\gamma \triangleq \frac{2c}{\sigma^2} + \xi$, where ξ is the unique positive root of the Lundberg fundamental equation

$$\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta = 0. \tag{2.4}$$

Gerber and Landry (1998) provide a probabilistic interpretation of $\Psi(u)$ and demonstrate that it satisfies the defective renewal equation

$$\begin{aligned}\Psi(u) &= \int_0^u \Psi(u-x)g(x)\mathrm{d}x + \omega(0)\mathrm{e}^{-\gamma u} + \int_u^\infty \omega(u-x)g(x)\mathrm{d}x \\ &\quad - \mathrm{e}^{-\gamma u}\int_0^\infty \omega(-x)g(x)\mathrm{d}x,\end{aligned} \tag{2.5}$$

for $u \geq 0$. Here, $g(x) = (g_1 * g_2)(x) \triangleq \int_0^x g_1(y)g_2(x-y)\mathrm{d}y$ with

$$g_1(x) = \gamma\mathrm{e}^{-\gamma x}, \quad \text{and} \quad g_2(x) = \frac{2\lambda}{\sigma^2\gamma}\mathrm{e}^{\xi x}\int_x^\infty \mathrm{e}^{-\xi y}p(y)\mathrm{d}y. \tag{2.6}$$

Lin and Willmot (1999) provide another representation in terms of compound geometric tails. They first denote

$$G(u) \triangleq \int_0^u g(x)\mathrm{d}x \quad \text{and} \quad K(u) \triangleq q\sum_{n=0}^{\infty} G^{*n}(u), \quad u \geq 0, \tag{2.7}$$

where $q = 1 - G(\infty) = \delta/\left(\frac{\sigma^2}{2}\xi^2 + c\xi\right)$. $G(u)$ is a defective distribution function while $K(u)$ is a compound geometric distribution function with probability mass q at 0. They demonstrate that

$$\Psi(u) = \frac{1}{q}\left\{\omega(0)\mathrm{e}^{-\gamma u} + \int_u^{\infty} \omega(u-x)g(x)\mathrm{d}x - \mathrm{e}^{-\gamma u}\int_0^{\infty} \omega(-x)g(x)\mathrm{d}x\right\} * K(u), \tag{2.8}$$

for $u \geq 0$.

According to Rolski et al. (1999), the family of phase-type distributions is dense in the class of all distributions defined on $[0, \infty)$ (see also Tijms (1994)). In addition, when the claim size distribution is phase-type, $K(u)$ has a rather simple representation. We now recall some known results on applications of phase-type distributions in ruin theory in the next subsection.

2.1. Phase-type claim size

Phase-type distributions play an important role in ruin theory. For more details (see Asmussen (2000) and Rolski et al. (1999)). In short, a phase-type distribution is the distribution of the first arrival to a single absorbing state in a finite state continuous time Markov chain. More specifically, consider a continuous-time Markov chain $\{J_t; t \geq 0\}$ with finite state space $\{1, 2, \ldots, k, \Xi\}$ where Ξ is the only absorbing state and all other states are transient. The intensity matrix of such a process can be written as

$$Q = \begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{0}^t & 0 \end{pmatrix}, \tag{2.9}$$

where $\mathbf{a} = -\mathbf{A}\mathbf{1}$ and $\mathbf{1} = (1, 1, \ldots, 1)^t$ is a column vector of 1's. Lemma 2.2.1 in Neuts (1981) implies that $\mathbf{A}$ is nonsingular and all entries of $(s\mathbf{I} - \mathbf{A})^{-1}$ are non-negative for any $s \geq 0$ according to Lemma 8.2.5 in Rolski et al. (1999).

Let the absorption time be $T \triangleq \inf\{t \geq 0 | J_t = \Xi\}$ and the initial distribution $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_k)$ (for states $1, \ldots, k$) and μ_Δ (for the absorbing state), then the distribution function of T is given by

$$F_T(t) = 1 - P(T > t) = 1 - \boldsymbol{\mu}\exp\{\mathbf{A}t\}\mathbf{1}, \quad t \geq 0. \tag{2.10}$$

The distribution of T is completely determined by the initial distribution $\boldsymbol{\mu}$ and sub-intensity matrix $\mathbf{A}$. T is said to be phase-type distributed and its distribution function is represented by $F_T \sim PH(\boldsymbol{\mu}, \mathbf{A})$.

Denote the L_1 norm of the initial distribution $\boldsymbol{\mu}$ by $\|\boldsymbol{\mu}\| = \sum_{i=1}^k \mu_i$. If $\|\boldsymbol{\mu}\| = 1$, F_T is a continuous distribution function; otherwise, F_T is a zero-modified probability distribution function, then the corresponding p.d.f. is

$$f_T(t) = \begin{cases} \boldsymbol{\mu}\mathrm{e}^{\mathbf{A}t}\mathbf{a}, & t > 0, \\ (1 - \|\boldsymbol{\mu}\|)\delta_0(t), & t = 0, \end{cases}$$

where δ_0 is a Dirac delta function centered at zero. In some cases, we need to extend the definition of phase-type distribution to that of the defective phase-type distribution. The defective phase-type distribution has p.d.f. $f_T(t) = \boldsymbol{\mu}\mathrm{e}^{\mathbf{A}t}\mathbf{a}$ for $t > 0$ and $F_T(\infty) = \|\boldsymbol{\mu}\| < 1$.

Phase-type distributions are closed under convolution operations. This closedness property is crucial for simplifying Eq. (2.8) when the claim size distribution is phase-type. Two key related results are provided by Ren (2005) and Neuts (1981) and are stated in Appendix A without proof. It is necessary to point out the distribution of $K(u)$ below is also given by Theorem 2.1 of Chapter VIII in Asmussen (2000) and Lemma 2.2 in Ren (2005). Here, we provide an alternative new proof.

Lemma 2.1. *Let claim size distribution be phase-type with representation $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$. Denote the vector $\boldsymbol{\nu} \triangleq \left(2\lambda/\sigma^2\gamma\right)\boldsymbol{\mu}(\xi\mathbf{I} - \mathbf{A})^{-1}$, then $\boldsymbol{\nu}$ is a non-negative vector and $\|\boldsymbol{\nu}\| < 1$. Furthermore, $G(u)$ defined in Eq. (2.7) is a defective phase-type distribution function with representation $PH(\boldsymbol{\vartheta}, \mathbf{B})$, where*

$$\boldsymbol{\vartheta} = (\boldsymbol{\nu}, 0), \qquad \mathbf{B} = \begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{0}^t & -\gamma \end{pmatrix}. \tag{2.11}$$

In this case, $K(u)$ is a zero-modified phase-type distribution function with representation $(\boldsymbol{\vartheta}, \mathbf{V})$, where $\mathbf{V} = \mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}$ is a non-singular sub-intensity matrix and $\mathbf{b} = -\mathbf{B}\mathbf{1}$.

Proof. Since $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, then the Lundberg fundamental equation (2.4) can be rewritten as

$$\frac{\sigma^2}{2}\xi^2 + c\xi - \lambda\xi\boldsymbol{\mu}(\xi\mathbf{I} - \mathbf{A})^{-1}\mathbf{1} - \delta = 0. \tag{2.12}$$

So $\|\boldsymbol{\nu}\| = \boldsymbol{\nu}\mathbf{1} = 1 - \delta/\left(\frac{\sigma^2}{2}\xi^2 + c\xi\right) = 1 - q < 1$.

In this case, $g_2(x) = \boldsymbol{\nu}\mathrm{e}^{\mathbf{A}x}\mathbf{a}$ is the p.d.f. of a defective phase-type distribution with representation $PH(\boldsymbol{\nu}, \mathbf{A})$. Since $g(x) = \int_0^x g_1(x-y)g_2(y)\mathrm{d}y$, Lemma A.1 implies $G(u)$ is a defective phase-type distribution with representation $PH(\boldsymbol{\vartheta}, \mathbf{B})$.

According to the definition of $K(u)$, for $s \geq 0$

$$\int_{0-}^{\infty} \mathrm{e}^{-su}\mathrm{d}K(u) = \frac{q}{1 - \widetilde{g}(s)} = \frac{1 - \|\boldsymbol{\vartheta}\|}{1 - \boldsymbol{\vartheta}(s\mathbf{I} - \mathbf{B})^{-1}\mathbf{b}} = (1 - \|\boldsymbol{\vartheta}\|)(1 + \boldsymbol{\vartheta}(s\mathbf{I} - (\mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}))^{-1}\mathbf{b}),$$

since $\boldsymbol{\vartheta}(s\mathbf{I} - (\mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}))^{-1}\mathbf{b} - \boldsymbol{\vartheta}(s\mathbf{I} - \mathbf{B})^{-1}\mathbf{b} = \boldsymbol{\vartheta}(s\mathbf{I} - \mathbf{B})^{-1}\mathbf{b}\boldsymbol{\vartheta}(s\mathbf{I} - (\mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}))^{-1}\mathbf{b}$. Finally, the result follows by observing that $(1 - \|\boldsymbol{\vartheta}\|)(1 + \boldsymbol{\vartheta}(s\mathbf{I} - (\mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}))^{-1}\mathbf{b})$ is the Laplace transform of a zero-modified phase-type distribution function with representation $(\boldsymbol{\vartheta}, \mathbf{B} + \mathbf{b}\boldsymbol{\vartheta})$. □

We now move on to the main result of this subsection—the explicit matrix-type representation of the EDP function. First denote

$$\omega_1 = \omega(0) - \boldsymbol{\nu}_\omega\mathbf{1}, \qquad \boldsymbol{\nu}_\omega = -\int_0^{\infty} \omega(-l)\boldsymbol{\vartheta}\mathrm{e}^{\mathbf{B}l}\mathbf{B}\mathrm{d}l$$

and

$$\boldsymbol{\alpha}_1 = [1, \mathbf{0}^t], \qquad \mathbf{T}_1 = \begin{pmatrix} -\gamma & \gamma\boldsymbol{\vartheta} \\ \mathbf{0}^t & \mathbf{V} \end{pmatrix}. \tag{2.13}$$

Theorem 2.1. *When the claim size distribution is phase-type with representation $PH(\boldsymbol{\mu}, \mathbf{A})$, then*

$$\Psi(u) = \frac{1}{q}\left[(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})\mathrm{e}^{\mathbf{V}u}\mathbf{1} + \omega_1\boldsymbol{\alpha}_1\mathrm{e}^{\mathbf{T}_1 u}\mathbf{1}\right], \quad u \geq 0. \tag{2.14}$$

Proof. When the claim size is phase-type distributed with representation $PH(\boldsymbol{\mu}, \mathbf{A})$, then

$$\int_u^{\infty} \omega(u-x)g(x)\mathrm{d}x = \left[-\boldsymbol{\vartheta}\int_0^{\infty} \omega(-l)\mathrm{e}^{\mathbf{B}l}\mathbf{B}\,\mathrm{d}l\right]\mathrm{e}^{\mathbf{B}u}\mathbf{1} = \boldsymbol{\nu}_\omega\mathrm{e}^{\mathbf{B}u}\mathbf{1}, \quad u \geq 0. \tag{2.15}$$

It is important to point out that $\boldsymbol{\nu}_\omega$ may not be non-negative. However, $\boldsymbol{\nu}_\omega$ can be decomposed into two non-negative vectors $\boldsymbol{\nu}_+$, $\boldsymbol{\nu}_-$, i.e.

$$\boldsymbol{\nu}_\omega = \boldsymbol{\nu}_+ - \boldsymbol{\nu}_-. \tag{2.16}$$

Eq. (2.8) can be reduced to

$$\Psi(u) = \frac{1}{q}\left[\boldsymbol{\nu}_+ e^{\mathbf{B}u}\mathbf{1} - \boldsymbol{\nu}_- e^{\mathbf{B}u}\mathbf{1} + \omega_1 e^{-\gamma u}\right] * K(u), \quad u \geq 0. \tag{2.17}$$

Let $H_1(t)$, $H_+(t)$ and $H_-(t)$ be phase-type distribution functions with representations $PH(\boldsymbol{\alpha}_1, \mathbf{T}_1)$, $PH(\boldsymbol{\nu}_+, \mathbf{V})$ and $PH(\boldsymbol{\nu}_-, \mathbf{V})$ respectively, then

$$\begin{aligned}\Psi(u) &= \frac{\omega_1}{q}\left[K(u) - (1 - e^{-\gamma u}) * K(u)\right] \\ &\quad + \frac{1}{q}[\boldsymbol{\nu}_+ e^{\mathbf{B}u}\mathbf{1}] * K(u) - \frac{1}{q}[\boldsymbol{\nu}_- e^{\mathbf{B}u}\mathbf{1}] * K(u) \\ &= \frac{\omega_1}{q}\left(K(u) - H_1(u)\right) + \frac{1}{q}\left(\boldsymbol{\nu}_\omega \mathbf{1} K(u) - H_+(u) + H_-(u)\right) \\ &= \frac{1}{q}\left[\omega(0)K(u) - \omega_1 H_1(u) - H_+(u) + H_-(u)\right], \quad u \geq 0\end{aligned}$$

where the second equality is obtained by Lemma A.2. Straightforward simplifications reduce $\Psi(u)$ above to Eq. (2.14).

An interesting special case is when the penalty function $\omega(x) = 1$ for $x \leq 0$. This corresponds to the Laplace transform of the ruin time. For this case, we have $\boldsymbol{\nu}_\omega = \boldsymbol{\vartheta}$, $\omega_1 = q$ and Eq. (2.14) simplifies to

$$\Psi(u) = E[e^{-\delta\widetilde{\tau}}\mathbb{I}(\widetilde{\tau} < \infty)|U_0 = u] = \boldsymbol{\alpha}_1 e^{\mathbf{T}_1 u}\mathbf{1}, \quad u \geq 0. \tag{2.18}$$

Remark 2.1. The above theorem extends Ren's (2005) result. He provides separate matrix-type expressions for the probability of ruin, the Laplace transform of the time of ruin, the expected value of the time of ruin and the discounted moments of the deficit at ruin. Our result combines all of these results through the EDP function and provides an analytical expression for the joint distribution of the time of ruin and the deficit at ruin.

3. Model formulation

An insurer's surplus is typically invested in the financial markets where it is exposed to all of the usual risk factors and therefore prone to market fluctuations. As noted earlier, the volatility itself is observed to fluctuate through time and must be modeled as a risk factor of its own. To this end, we assume the volatility follows a stochastic process $\{\sigma_t; t \geq 0\}$, and the surplus process $\{U_t; t \geq 0\}$ satisfies the SDE

$$dU_t = c\,dt - dS_t + \sigma_t dW_t, \quad t \geq 0 \quad \text{and} \quad U_0 = u. \tag{3.1}$$

Following Fouque et al. (2000), we assume that the volatility is driven by an OU process Y_t such that $\sigma_t = f(Y_t)$ for some positive function f. As such, Y_t is defined by

$$dY_t = \alpha(m - Y_t)dt + \beta\, dZ_t, \quad Y_0 = y_0, \tag{3.2}$$

where $\alpha > 0$ is the rate of mean reversion, m is the long-run mean level of Y_t, $\{Z_t; t \geq 0\}$ is a standard Brownian motion with $d\langle W, Z\rangle_t = \rho dt$ for some correlation coefficient $\rho \in [-1, 1]$ and independent of the aggregate claims process $\{S_t; t \geq 0\}$. As noted by Heston (1993), the correlation between volatility and price is necessary to capture the so called leverage effect—the tendency of volatility to increase as prices drop and decrease as prices rise. The same argument applies to insurance surplus. In this work, we therefore assume that the correlation coefficient ρ is nonzero.

In Fig. 3.1, we have simulated a sample path for the surplus process under constant volatility and stochastic volatility using the same underlying Brownian motion and jump paths. The generated volatility path is also shown. In this example, the stochastic volatility function is $f(y) = \exp\{y\}$ and the constant volatility is $\sigma^2 = \langle f^2\rangle$, corresponding to the average of the volatility function under the invariant distribution for the OU process (see Section 3.1 for more details). Although the stochastic volatility model was chosen to have the same mean behavior as the constant volatility case, the resulting SV surplus sample path clearly shows more extreme, almost jump-like behavior between claim arrivals. In fact, there are both upward and downward jump-like features, which is precisely what is observed in real return data.

As usual, the time of (ultimate) ruin is defined by $\tau = \inf\{t \geq 0 : U_t \leq 0\}$ and the EDP function is given by

$$\Phi(u, y) = \mathbb{E}[e^{-\delta\tau}\omega(U_\tau)\,\mathbb{I}(\tau < \infty)|U_0 = u, Y_0 = y]. \tag{3.3}$$

Assumption 3.1. The EDP function $\Phi(u, y) \in C^{2,2}(\mathbb{R}_+ \times \mathbb{R})$ when constrained on $\mathbb{R}_+ \times \mathbb{R}$.[1]

Under the above modeling assumptions, we will show that $\Phi(u, y)$ satisfies a non-linear IDE in Section 4. This IDE unfortunately does not have an explicit solution. However, when the driving process $\{Y_t; t \geq 0\}$ is fast mean-reverting with a finite variance, i.e. $\alpha \triangleq \frac{1}{\epsilon}$ where $\epsilon \ll 1$, we develop an asymptotic expansion of $\Phi(u, y)$ and determine the first few terms.

To engage in this program, $\Phi(u, y)$ is expanded in powers of $\sqrt{\epsilon}$ as follows

$$\begin{aligned}\Phi(u, y) &= \Phi_0(u, y) + \sqrt{\epsilon}\Phi_1(u, y) + \epsilon\Phi_2(u, y) \\ &\quad + \epsilon\sqrt{\epsilon}\Phi_3(u, y) + \cdots,\end{aligned} \tag{3.4}$$

where $\Phi_0(u, y)$, $\Phi_1(u, y)$, $\Phi_2(u, y)$, $\Phi_3(u, y)$ have boundary conditions $\Phi_0(u, y) = \omega(u)$, $\Phi_1(u, y) = 0$, $\Phi_2(u, y) = \Phi_2(0, y)$ and $\Phi_3(u, y) = \Phi_3(0, y)$ respectively for $u \leq 0$. Furthermore, we assume $\Phi_{0,1,2,3}(u, y) \in C^{2,2}(\mathbb{R}_+ \times \mathbb{R})$ when constrained on $\mathbb{R}_+ \times \mathbb{R}$.

Several crucial properties of OU processes which are necessary for deriving the IDE for $\Phi(u, y)$ and determining the approximation terms Φ_0 and Φ_1 are explored in the next subsection.

3.1. Properties of OU processes

This subsection introduces some background necessary to understand singular perturbation theory. For more details, see Fouque et al. (2000) and references therein. We begin with a quick review of properties of OU processes.

The OU process $\{Y_t; t \geq 0\}$ in Eq. (3.2) is a strong Markov process with infinitesimal generator

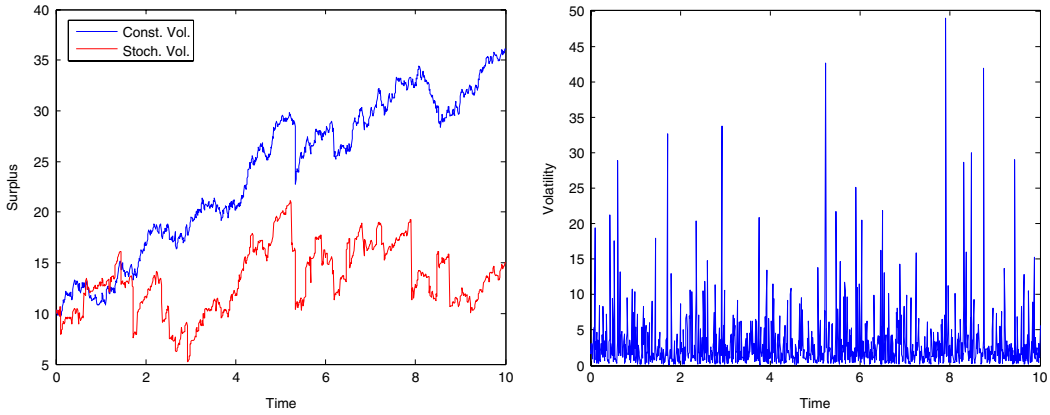
$$\mathcal{L}_{OU} = \frac{\beta^2}{2}\frac{\partial^2}{\partial y^2} + \alpha(m - y)\frac{\partial}{\partial y}.$$

[1] In ruin theory, many authors derive the IDEs for the EDP function under the implicit assumption that EDP function is twice continuously differentiable. Theoretically, discussion of the differentiability of EDP function is necessary. Under the perturbed compound Poisson model with continuous claim density $p(y)$, Cai and Yang (2005) use stochastic control techniques to show that the ruin probability is twice continuously differentiable. That result is extended to the EDP function with bounded penalty by Chen et al. (2007). Furthermore, when the surplus process $\{U_t\}_{t\geq 0}$ follows a one-dimensional Lévy process, satisfying the ACP condition, with bounded characteristic exponent ψ and penalty function ω, then Boyarchenko and Levendorskiĭ (2002) shows the corresponding EDP function Ψ satisfies

$$\langle \Psi, (\delta - \mathscr{H})M\rangle_{L_2} = 0, \quad \forall M \in C_0^\infty((0, \infty)).$$

Here, $\mathscr{H}$ denotes the generator $\mathscr{H} \triangleq -\psi(i\frac{\partial}{\partial x})$, where $i = \sqrt{-1}$. For more details, see Boyarchenko and Levendorskiĭ (2002).

For the SV extended risk model defined in Eq. (3.1), it is difficult to demonstrate the differentiability of the EDP function $\Phi(u, y)$ by traditional techniques in ruin theory. However, we conjecture that Assumption 3.1 is valid.



(a) Surplus paths. (b) Volatility path.

Fig. 3.1. (a) Sample paths of the surplus process under constant and stochastic volatility. (b) Sample path of the volatility process.

For fixed time t, if Y_t is of invariant distribution with p.d.f. $n(y)$, then $\mathbb{E}h(Y_t) = \mathbb{E}h(Y_s)$, for any $s \geq t$ and any smooth function h, when the expectations exist. Consequently, $\mathbb{E}[\mathcal{L}_{OU}h(Y_t)] = 0$. Furthermore, let $\mathcal{L}^*_{OU}$ denote the adjoint operator of $\mathcal{L}_{OU}$, then

$$\int_{-\infty}^{\infty} \mathcal{L}^*_{OU} n(y)h(y)\mathrm{d}y = \int_{-\infty}^{\infty} n(y)\mathcal{L}_{OU}h(y)\mathrm{d}y = 0, \tag{3.5}$$

implying the adjoint equation $\mathcal{L}^*_{OU}n(y) = 0$. Fouque et al. (2000) show that

$$n(y) = \frac{1}{\sqrt{2\pi\, v^2}} \exp\left\{-\frac{(y-m)^2}{2v^2}\right\}, \quad y \in (-\infty, \infty), \tag{3.6}$$

where $v^2 \triangleq \beta^2/(2\alpha)$ solves the adjoint equation. Therefore, the steady state of the OU process is normally distributed with mean m and standard deviation v.

In addition, the unique solution to the SDE in Eq. (3.2) is well known to be

$$Y_t = m + (y_0 - m)\mathrm{e}^{-\alpha t} + \beta \int_0^t \mathrm{e}^{-\alpha(t-s)}\mathrm{d}Z_s, \quad t \geq 0, \tag{3.7}$$

then

$$Y_t \sim N\left(m + (y_0 - m)\mathrm{e}^{-\alpha t}, v^2(1 - \mathrm{e}^{-2\alpha t})\right), \quad t \geq 0,$$

where N (mean, variance) represents the normal distribution function with given mean and variance. With v held finite and fixed, as $\alpha \to \infty$ or $t \to \infty$, Y_t weakly converges to the invariant distribution.

The averaging operator with respect to the invariant distribution of the OU process appears several times in our analysis. Denote this average by $\langle\cdot\rangle$, specifically

$$\langle h\rangle \triangleq \int_{-\infty}^{\infty} n(y)h(y)\mathrm{d}y, \tag{3.8}$$

for any function $h(y)$. This averaging operation is called *centering*. Centering naturally appears in the solvability conditions for the *Poisson Equation*

$$\mathcal{L}_0\chi(y) + k(y) = 0. \tag{3.9}$$

Here, the operator $\mathcal{L}_0$ is defined as

$$\mathcal{L}_0 \triangleq \frac{1}{\alpha}\mathcal{L}_{OU} = v^2\frac{\partial^2}{\partial y^2} + (m-y)\frac{\partial}{\partial y} \tag{3.10}$$

and $k(y)$ is a source term for the Poisson Equation. Eq. (3.9) cannot be solved for arbitrary $k(y)$, rather there is a strict integrability condition on the source. To see this, let $\mathcal{L}^*_0 = \frac{1}{\alpha}\mathcal{L}^*_{OU} = v^2\frac{\partial^2}{\partial y^2} - \frac{\partial}{\partial y}((m-y)\cdot)$ be the adjoint operator of $\mathcal{L}_0$. Since

$$\begin{aligned}\langle k\rangle &= -\langle\mathcal{L}_0\chi\rangle = -\int_{-\infty}^{\infty}\mathcal{L}_0\chi(y)n(y)\mathrm{d}y \\ &= -\int_{-\infty}^{\infty}\chi(y)\mathcal{L}^*_0 n(y)\mathrm{d}y = 0,\end{aligned} \tag{3.11}$$

to ensure the equation has bounded solutions, the Poisson Equation must satisfy the so called *Centering Condition*: $\langle k\rangle = 0$. In other words, there exists a function $\hat{k}(y)$ such that $k(y)$ can be rewritten as $k(y) = \hat{k}(y) - \langle\hat{k}\rangle$.

Given a source satisfying the centering condition, solutions of the Poisson Equation inherit growth properties from the source. When $|k(y)| \leq C_1(1+|y|^n)$ for some constant C_1 and non-negative integer n, then

$$|\chi(y)| \leq \begin{cases} C_2(1+|y|^n), & n > 0 \\ C_2(1+log(1+|y|)), & n = 0, \end{cases} \tag{3.12}$$

for some constant C_2. These growth properties are very useful in establishing bounds on the truncation error in the asymptotic expansion.

The final property which will prove to be very useful is the characterization of the null space of the generator $\mathcal{L}_0$. Fouque et al. (2000) demonstrate that the general solution of the ODE

$$\mathcal{L}_0 h(y) = v^2 h''(y) + (m-y)h'(y) = 0 \tag{3.13}$$

is $h(y) = C_1\int_{-\infty}^{y}\exp\{(m-s)^2/2v^2\}\,\mathrm{d}s + C_2$, for constants C_1 and C_2.

Remark 3.1. Since $\int_{-\infty}^{\infty}\exp\{(m-s)^2/2v^2\}\mathrm{d}s = \infty$, well-behaved solutions of (3.13) at infinity must have $C_1 = 0$. Therefore, the well behaved null space of the operator $\mathcal{L}_0$ contains only constant functions.

This completes the collection of main results required to carry through with the asymptotic expansion of the EDP function. We now proceed to derive the IDE which the EDP function satisfies.

4. Associated integro-differential equation

We work on the probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$, where the information filtration $\mathcal{F}_t$ is the augmentation of the natural filtration of (U_t, Y_t). Without loss of generality, $\{\mathcal{F}_t\}_{t\geq 0}$ is assumed right-continuous. Under the filtration $\{\mathcal{F}_t\}_{t\geq 0}$, $(U_t, Y_t)_{t\geq 0}$ is a strong Markov process and the ruin time τ is a stopping time.

The multi-dimensional Itô formula (see e.g. Øksendal and Sulem (2005)) implies the infinitesimal generator $\mathcal{L}$ of (U_t, Y_t) acts on arbitrary functions $h \in C^{2,2}(\mathbb{R}^2)$ as follows:

$$\begin{aligned}\mathcal{L}h(u,y) &= \frac{f^2(y)}{2}\frac{\partial^2 h(u,y)}{\partial u^2} + c\frac{\partial h(u,y)}{\partial u} \\ &\quad + \lambda\int_0^\infty h(u-z,y)p(z)\mathrm{d}z - \lambda h(u,y) + \frac{\beta^2}{2}\frac{\partial^2 h(u,y)}{\partial y^2} \\ &\quad + \rho\beta f(y)\frac{\partial^2 h(u,y)}{\partial u\partial y} + \alpha(m-y)\frac{\partial h(u,y)}{\partial y}.\end{aligned}$$

We now state the main result of this section.

Proposition 4.1. *Under Assumption 3.1, $\Phi(u,y)$ satisfies the integro-differential equation*

$$\begin{cases}(\mathcal{L}-\delta)\Phi(u,y) = 0, & u > 0 \\ \Phi(u,y) = \omega(u), & u \le 0.\end{cases} \tag{4.1}$$

Proof. When $u \le 0$, $\tau = 0$, then the EDP function in Eq. (3.3) implies

$$\Phi(u,y) = \omega(u). \tag{4.2}$$

When $u > 0$, a modified version of Theorem 2.4 in Chen et al. (2007) suffices which we now prove.

Firstly, let $H_t \triangleq \mathrm{e}^{-\delta t}\Phi(U_t, Y_t)$, we will prove that $\{H_{\tau\wedge t}; t \ge 0\}$ is a $(\mathbb{P}_{u,y}, \mathscr{F}_{\tau\wedge t})$ martingale, and in particular a Doob martingale, by showing the following equality holds:

$$H_{\tau\wedge t} = \mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)|\mathscr{F}_{\tau\wedge t}]. \tag{4.3}$$

On the set $\{\tau \le t\}$, since $U_\tau \le 0$, Eq. (4.2) implies

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{-\delta\tau}\,\omega(U_\tau)|\mathscr{F}_{\tau\wedge t}]\,\mathbb{I}(\tau \le t) &= \mathrm{e}^{-\delta\tau}\,\omega(U_\tau)\,\mathbb{I}(\tau \le t) \\ &= H_\tau\,\mathbb{I}(\tau \le t).\end{aligned}$$

On the set $\{\tau > t\}$, the strong Markov property of $\{(U_t, Y_t)\}_{t\ge 0}$ implies

$$\begin{aligned}&\mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)|\mathscr{F}_{\tau\wedge t}]\,\mathbb{I}(\tau > t) = \mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)|\mathscr{F}_t]\,\mathbb{I}(\tau > t) \\ &\quad = \mathrm{e}^{-\delta t}\mathbb{E}[\mathrm{e}^{-\delta\tau^*}\omega(U_{\tau^*})|U_t, Y_t]\mathbb{I}(\tau > t) = H_t\,\mathbb{I}(\tau > t),\end{aligned}$$

where τ^* is the ruin time of the risk process $\{U_{t+s}; s \ge 0\}$. These two computations imply that Eq. (4.3) holds.

Secondly, we will localize $\Phi(U_t, Y_t)$ through the stopping time $\tau-$. By Assumption 3.1, Φ is twice continuously differentiable, therefore Dynkin's formula (see e.g. Rolski et al. (1999)) implies for any stopping time $\hat\tau < \tau$

$$\begin{aligned}&\mathbb{E}\left[H_{\hat\tau\wedge t} \mid U_0 = u, Y_0 = y\right] - \Phi(u,y) \\ &\quad = \mathbb{E}\left[\int_0^{\hat\tau\wedge t} \mathrm{e}^{-\delta s}(\mathcal{L}-\delta)\,\Phi(U_s, Y_s)\mathrm{d}s \,\middle|\, U_0 = u, Y_0 = y\right].\end{aligned} \tag{4.4}$$

Since $H_{\tau\wedge t}$ has already been proven a $(\mathbb{P}_{u,y}, \mathscr{F}_{\tau\wedge t})$ martingale, then

$$\mathbb{E}\left[\int_0^t \mathbb{I}(\hat\tau > s)\,\mathrm{e}^{-\delta s}(\mathcal{L}-\delta)\Phi(U_s, Y_s)\mathrm{d}s \,\middle|\, U_0 = u, Y_0 = y\right] = 0.$$

Now take any $0 < \zeta < u$ and let $\hat\tau$ be the first exit time of the process (U_t, Y_t) from the open ball $B((u,y),\zeta)$ centered at (u,y) with radius ζ. By Assumption 3.1, (i) $\mathbb{P}_{u,y}(\hat\tau > 0) > 0$ and (ii) $(\mathcal{L}-\delta)\Phi(U_s, Y_s)$ is uniformly bounded on the closure of the ball $B((u,y),\zeta)$ whenever $s < \hat\tau$. Consequently,

$$\mathbb{E}\left[\mathbb{I}(\hat\tau > s)\,(\mathcal{L}-\delta)\Phi(U_s, Y_s)\,\middle|\, U_0 = u, Y_0 = y\right] = 0.$$

Taking $s \downarrow 0$ and appealing to the bounded convergence theorem leads to $(\mathcal{L}-\delta)\Phi(u,y) = 0$ and completes the proof. □

Remark 4.1. Proposition 4.1 is easily extended to account for penalties which depend on the state of volatility as well as the deficit at ruin, i.e. for $\omega(U_\tau, Y_\tau)$. Such penalties will simply require replacing the boundary condition in (4.1) with $\Phi(u,y) = \omega(u,y)$ for any $u \le 0$.

The new IDE appearing in Eq. (4.1) is significantly more complicated than the classical IDE in Eq. (2.2). The new IDE is firstly two dimensional (rather than one) and secondly contains nonlinearities in its coefficients which are inherited from the mapping of the driving OU process Y_t to volatility. To the best of our knowledge, no closed-form solution exists for this IDE. However, in the next section, through singular perturbation theory techniques, we will explicitly obtain the first few terms of the expansion (3.4) of $\Phi(u,y)$ in powers of the inverse mean-reversion rate ϵ.

5. Approximation of the EDP function

In this section, we obtain an asymptotic solution of the IDE (4.1) and explicitly construct the first few terms of the expansion. The expansion will be carried out in the homogenization limit where the mean-reversion rate α of the driving source of volatility Y_t is large while the variance of its invariant distribution is kept fixed at v^2. This is precisely the limit which Fouque et al. (2000) use in the context of equity derivatives.

5.1. Decoupling

Since $\Phi(u,y)$ is parameterized by the small parameter ϵ, we rewrite $\Phi(u,y)$ as $\Phi^\epsilon(u,y)$ to emphasize this dependence. Further, we rewrite the IDE (4.1) as

$$\begin{cases}\left(\frac{1}{\epsilon}\mathcal{L}_0 + \frac{1}{\sqrt{\epsilon}}\mathcal{L}_1 + \mathcal{L}_2\right)\Phi^\epsilon(u,y) = 0, & \text{for } u > 0, \\ \Phi^\epsilon(u,y) = \omega(u), & \text{for } u \le 0,\end{cases} \tag{5.1}$$

to highlight the scales of the various operators involved. In the above, two new operators are introduced

$$\mathcal{L}_1 h(u,y) \triangleq \sqrt{2}\rho v f(y)\frac{\partial^2 h(u,y)}{\partial u\partial y}, \tag{5.2}$$

and

$$\begin{aligned}\mathcal{L}_2 h(u,y) &\triangleq \frac{f^2(y)}{2}\frac{\partial^2 h(u,y)}{\partial u^2} + c\frac{\partial h(u,y)}{\partial u} \\ &\quad + \lambda\int_0^\infty h(u-z,y)p(z)\mathrm{d}z - (\lambda+\delta)h(u,y),\end{aligned} \tag{5.3}$$

for arbitrary $h \in C^{2,1}(\mathbb{R}_+ \times \mathbb{R})$. Notice that the operator $\mathcal{L}_2$ is closely related to the classical perturbed compound Poisson risk model's infinitesimal generator $\mathcal{L}_c$ in Eq. (2.3). Specifically, if the constant volatility σ in $\mathcal{L}_c$ is replaced with the volatility mapping function $f(y)$, the two differ only by the discounting rate: $\mathcal{L}_2 = \mathcal{L}_c(f(y)) - \delta$.

Now, plugging Eq. (3.4) into Eq. (5.1) and collecting terms of equal order in ϵ, we have

$$\begin{aligned}0 = (\mathcal{L}-\delta)\Phi^\epsilon(u,y) &= \frac{1}{\epsilon}\mathcal{L}_0\Phi_0 + \frac{1}{\sqrt{\epsilon}}(\mathcal{L}_0\Phi_1 + \mathcal{L}_1\Phi_0) \\ &\quad + (\mathcal{L}_0\Phi_2 + \mathcal{L}_1\Phi_1 + \mathcal{L}_2\Phi_0) \\ &\quad + \sqrt{\epsilon}(\mathcal{L}_0\Phi_3 + \mathcal{L}_1\Phi_2 + \mathcal{L}_2\Phi_1) + \cdots,\end{aligned}$$

for $u > 0$. Equating each coefficient to zero, the first four terms imply

$$\epsilon^{-1}\text{-order term:} \quad \mathcal{L}_0\Phi_0(u,y) = 0; \tag{5.4}$$

$$\epsilon^{-1/2}\text{-order term:} \quad \mathcal{L}_0\Phi_1(u,y) + \mathcal{L}_1\Phi_0(u,y) = 0; \tag{5.5}$$

$$\epsilon^{0}\text{-order term:} \quad \mathcal{L}_0\Phi_2(u,y) + \mathcal{L}_1\Phi_1(u,y) + \mathcal{L}_2\Phi_0(u,y) = 0; \tag{5.6}$$

$$\epsilon^{1/2}\text{-order term:} \quad \mathcal{L}_0\Phi_3(u,y) + \mathcal{L}_1\Phi_2(u,y) + \mathcal{L}_2\Phi_1(u,y) = 0. \tag{5.7}$$

Let us now analyze these individual equations in turn.

By Remark 3.1, well-behaved functions in the null space of the operator $\mathcal{L}_0$ are independent of y. Consequently, the ϵ^{-1}-order term implies $\Phi_0(u,y) = c_0(u)$ for some function c_0. Likewise, since $\mathcal{L}_1\Phi_0(u,y) = 0$, the $\epsilon^{-1/2}$-order term implies $\mathcal{L}_0\Phi_1(u,y) = 0$, therefore $\Phi_1(u,y) = c_1(u)$ for some function c_1.

The above two results imply that the lowest orders of the asymptotic expansion of the EDP function are independent of the current level of volatility. At first this is a surprising consequence. However, since volatility itself is fast mean-reverting with finite variance, no matter what its current level is, a short instant later it will be distributed according to its invariant distribution. This is the so-called homogenization effect.

Given that the first two terms are independent of y (i.e. $\Phi_{1,2}(u,y) = c_{1,2}(u)$) we rewrite $\Phi_{1,2}(u,y)$ as $\Phi_{1,2}(u)$. Now, the ϵ^0-order term (Eq. (5.6)) simplifies to

$$\mathcal{L}_0\Phi_2(u,y) + \mathcal{L}_2\Phi_0(u) = 0. \tag{5.8}$$

This is a Poisson Equation for $\Phi_2(u,y)$ with $\mathcal{L}_2\Phi_0(u)$ acting as a source. As discussed in Section 3.1, for the Poisson Equation to be well defined, the source term must satisfy the centering condition (3.11). In this context, the centering condition implies $\langle\mathcal{L}_2\Phi_0(u)\rangle = 0$. Rewriting in terms of the classical operator $\mathcal{L}_c$ we have

$$(\mathcal{L}_c(\langle f^2\rangle^{1/2}) - \delta)\Phi_0(u) = 0. \tag{5.9}$$

In the remainder of this article we set $\sigma^2 = \langle f^2\rangle$ for notational convenience. This equation for $\Phi_0(u)$ is precisely the IDE which the classical perturbed compound Poisson risk model's EDP function satisfies—see Eq. (2.2). Interestingly, our stochastic volatility extended risk model's EDP function is, in the limit as $\epsilon \downarrow 0$, identical to the classical EDP function with a variance equal to the average variance in the SV model. This is once again the effect of homogenization.

Now we move onto solving the Poisson Equation (5.8). Due to the centering condition, the source term for the Poisson Equation can be simplified considerably to

$$\mathcal{L}_2\Phi_0(u) = \mathcal{L}_2\Phi_0(u) - \langle\mathcal{L}_2\Phi_0(u)\rangle = \frac{1}{2}(f^2(y) - \sigma^2)\Phi_0''(u).$$

Substituting back into Eq. (5.8) yields

$$\begin{aligned}\Phi_2(u,y) &= -\frac{1}{2}\mathcal{L}_0^{-1}(f^2(y) - \sigma^2)\Phi_0''(u)\\ &= -\frac{1}{2}(\phi(y) + c_2(u))\Phi_0''(u).\end{aligned} \tag{5.10}$$

Here, $\mathcal{L}_0\phi(y) = f^2(y) - \sigma^2$ and $c_2(u)$ is an unknown function. Fouque et al. (2000) show through direct calculations that

$$\phi'(y) = \frac{1}{v^2 n(y)}\int_{-\infty}^{y}(f^2(z) - \sigma^2)\, n(z)\mathrm{d}z. \tag{5.11}$$

We now have the correction term $\Phi_2(u,y)$ but not $\Phi_1(u)$. For this, we must proceed to the next order and obtain a second centering condition.

The $\epsilon^{1/2}$-order term equation (5.7) is a Poisson Equation for $\Phi_3(u,y)$ with source term $\mathcal{L}_1\Phi_2(u,y) + \mathcal{L}_2\Phi_1(u)$. The centering condition therefore implies $\langle\mathcal{L}_1\Phi_2(u,y) + \mathcal{L}_2\Phi_1(u)\rangle = 0$. Upon substituting Eq. (5.10) into this equation and rewriting $\mathcal{L}_2$ in terms of $\mathcal{L}_c$, one finds

$$(\mathcal{L}_c(\sigma) - \delta)\Phi_1(u) = \frac{1}{2}\langle\mathcal{L}_1\phi(y)\rangle\,\Phi_0''(u) = \frac{\sqrt{2}}{2}\rho\, v\,\langle f\phi'\rangle\Phi_0'''(u). \tag{5.12}$$

This is very similar to the classical IDE (2.2); however, it contains an explicit source term which depends on the solution $\Phi_0(u)$.

Armed with the two IDEs (5.9) and (5.12), we are now in a position to provide a probabilistic interpretation of $\Phi_0(u)$ and $\Phi_1(u)$ of the expansion of the EDP function $\Phi(u,y)$.

Proposition 5.1. *The first two terms in the asymptotic expansion of the EDP function $\Phi(u,y)$ for the stochastic volatility extended risk model have natural representations in terms of the classical perturbed risk model. Specifically, let $\{U_t; t \geq 0\}$ be the process defined in Eq. (2.1) where the volatility is set to $\sigma^2 = \langle f^2(y)\rangle$, and let the corresponding ruin time be denoted $\widetilde{\tau}$, then*

$$\Phi_0(u) = \Psi(u) = \mathbb{E}\left[\mathrm{e}^{-\delta\widetilde{\tau}}\,\omega(U_{\widetilde{\tau}})\,\mathbb{I}(\widetilde{\tau} < \infty)\middle|\, U_0 = u\right]. \tag{5.13}$$

When $\int_{0-}^{\infty}\mathrm{e}^{-\xi u}\Phi_0'''(u)\mathrm{d}u < \infty$ and the Laplace transform $\widetilde{\Phi_1}(\xi) < \infty$,

$$\Phi_1(u) = -\frac{\sqrt{2}}{2}\,\rho\, v\,\langle f\phi'\rangle\,\mathbb{E}\left[\int_0^{\widetilde{\tau}}\mathrm{e}^{-\delta s}\,\Phi_0'''(U_s)\mathrm{d}s\middle|\, U_0 = u\right]. \tag{5.14}$$

Proof. According to known results on the perturbed compound Poisson risk model in Section 2, the IDE (5.9) implies Eq. (5.13).

Denote the right-hand side of Eq. (5.14) by $R(u)$. When $u \leq 0$, $\widetilde{\tau} = 0$, then $R(u) = 0$. For $u > 0$, take $\Delta t \ll 1$, then

$$\begin{aligned}R(u) =\ & (1 - \lambda\Delta t)(1 - \delta\Delta t)\mathbb{E}[R(u + c\Delta t + \sigma W(\Delta t))]\\ &+ \lambda\Delta t\int_0^u R(u - y)p(y)\mathrm{d}y\\ &- \mathbb{E}\left[\frac{\sqrt{2}}{2}\,\rho\, v\,\langle f\phi'\rangle\int_0^{\Delta t}\mathrm{e}^{-\delta s}\Phi_0'''(U_s)\mathrm{d}s\middle|\, U_0 = u\right] + o(\Delta t),\end{aligned}$$

where $o(\Delta t)$ represents a term with $\lim_{\Delta t\downarrow 0}\frac{o(\Delta t)}{\Delta t} = 0$.

Since $\mathbb{E}[R(u + c\Delta t + \sigma W(\Delta t))] - R(u) = c\,R'(u)\,\Delta t + \frac{\sigma^2}{2}R''(u)\,\Delta t + o(\Delta t)$, then as $\Delta t \downarrow 0$,

$$(\mathcal{L}_c(\sigma) - \delta)\,R(u) = \frac{\sqrt{2}}{2}\,\rho\, v\,\langle f\phi'\rangle\,\Phi_0'''(u). \tag{5.15}$$

Since $\Phi_1(u)$ and $R(u)$ satisfy the same IDE and have identical boundary conditions, and the solution of the IDE (5.12) is shown to be unique in Lemma 5.1 below, then $\Phi_1(u) = R(u)$. □

The above Proposition shows that $\Phi_0(u) = \Psi(u)$. Gerber and Landry (1998) demonstrate that the EDP function $\Psi(u)$ satisfies the defective renewal equation (2.5) by analyzing when ruin is caused by oscillation or by a claim separately. In our case, since the IDEs (5.9) and (5.12) which the SV correction terms Φ_0 and Φ_1 satisfy are quite similar – one contains a source term and the other does not – we choose to unite them into a single IDE and use Laplace transform to obtain explicit solutions.

Denote the Laplace transform of a function $f(x)$ defined on $[0,\infty)$ by

$$(\mathscr{A}f)(s) = \widetilde{f}(s) = \int_{0-}^{\infty}\mathrm{e}^{-sx}f(x)\mathrm{d}x, \quad s > 0 \tag{5.16}$$

and the inverse Laplace transform by $(\mathscr{A}^{-1}\widetilde{f})(x) = f(x)$.

Lemma 5.1. *Suppose that a function $R(u)$ satisfies the integro-differential equation*

$$\begin{cases}(\mathcal{L}_c(\sigma) - \delta)R(u) = v_1(u), & u > 0\\ R(u) = v_2(u), & u \leq 0.\end{cases} \tag{5.17}$$

Denote $Q(u) = \int_u^{\infty}v_2(u - z)p(z)\mathrm{d}z$ for $u \geq 0$. If $\widetilde{v_1}(\xi) < \infty$ and $\widetilde{Q}(\xi) < \infty$, where ξ is the positive solution of the Lundberg fundamental equation (2.4), then there exists a solution $R(u)$ such that $\widetilde{R}(\xi) < \infty$ given by

$$R(u) = \frac{1}{q}\left[v_2(0)\mathrm{e}^{-\gamma u} + \int_u^{\infty}v_2(u - z)g(z)\mathrm{d}z\right.$$

$$- \mathrm{e}^{-\gamma u}\int_0^\infty v_2(-z)g(z)\mathrm{d}z$$
$$- \frac{2}{\sigma^2}\mathrm{e}^{-\gamma u}\int_0^u \mathrm{e}^{(\gamma+\xi)z}\int_z^\infty v_1(x)\mathrm{e}^{-\xi x}\mathrm{d}x\mathrm{d}z\Bigg] * K(u), \quad (5.18)$$

for $u \geq 0$. Here, $K(u)$ is defined in Eq. (2.7) and $g(u) = (g_1 * g_2)(u)$ where $g_1(u)$ and $g_2(u)$ are defined in Eq. (2.6).

Proof. Taking Laplace transforms of Eq. (5.17) for $u > 0$, we have

$$\left[\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta\right]\widetilde{R}(s)$$
$$= \frac{\sigma^2}{2}(s\,v_2(0) + R'(0)) + c\,v_2(0) + \widetilde{v}_1(s) - \lambda\,\widetilde{Q}(s). \quad (5.19)$$

Since $\widetilde{R}(\xi) < \infty$, $\widetilde{v}_1(\xi) < \infty$ and $\widetilde{Q}(\xi) < \infty$, Lundberg fundamental equation (2.4) implies that

$$\frac{\sigma^2}{2}\left(\xi\,v_2(0) + R'(0)\right) + c\,v_2(0) + \widetilde{v}_1(\xi) - \lambda\,\widetilde{Q}(\xi) = 0. \quad (5.20)$$

Plugging the above into Eq. (5.19), and appealing to Lemma A.3 for the singleness of ξ, implies that

$$\widetilde{R}(s) = \frac{\frac{\sigma^2}{2}v_2(0)(s-\xi) + (\widetilde{v}_1(s) - \widetilde{v}_1(\xi)) - \lambda(\widetilde{Q}(s) - \widetilde{Q}(\xi))}{\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta}$$
$$= J_1(s) \times J_2(s), \quad (5.21)$$

where

$$J_1(s) = \frac{\frac{\sigma^2}{2}(s-\xi)(s+\gamma)}{\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta} \quad \text{and}$$
$$J_2(s) = \frac{v_2(0)}{s+\gamma} + \frac{\widetilde{v}_1(s) - \widetilde{v}_1(\xi)}{\frac{\sigma^2}{2}(s-\xi)(s+\gamma)} - \lambda\frac{\widetilde{Q}(s) - \widetilde{Q}(\xi)}{\frac{\sigma^2}{2}(s-\xi)(s+\gamma)}.$$

We now obtain the inverse Laplace transforms of J_1 and J_2.

Firstly, using the definition of $g_1(u)$ and $g_2(u)$ in Eq. (2.6) we have

$$\widetilde{g}(s) = \widetilde{g_1}(s)\widetilde{g_2}(s) = \frac{2\lambda}{\sigma^2}\frac{1}{\gamma+s}\int_0^\infty \mathrm{e}^{-(s-\xi)x}\int_x^\infty p(y)\mathrm{e}^{-\xi y}\mathrm{d}y\mathrm{d}x$$
$$= \frac{2\lambda(\widetilde{p}(\xi) - \widetilde{p}(s))}{\sigma^2(s+\gamma)(s-\xi)}.$$

Further, using the definition of $K(u)$ in Eq. (2.7), together with the above result, we have

$$\frac{1}{q}\int_{0-}^\infty \mathrm{e}^{-su}\mathrm{d}K(u) = \frac{1}{1-\widetilde{g}(s)}$$
$$= \frac{\frac{\sigma^2}{2}(s+\gamma)(s-\xi)}{\frac{\sigma^2}{2}(s+\gamma)(s-\xi) - \lambda(\widetilde{p}(\xi) - \widetilde{p}(s))} = J_1(s),$$

where the last equality follows from Eq. (2.4). Therefore,

$$(\mathscr{A}^{-1}J_1)(x) = \frac{1}{q}K'(x), \quad x \geq 0. \quad (5.22)$$

Secondly, it is well-known that

$$\left(\mathscr{A}^{-1}\left[\frac{\widetilde{f}(s) - \widetilde{f}(\xi)}{s-\xi}\right]\right)(x) = -\mathrm{e}^{\xi x}\int_x^\infty \mathrm{e}^{-\xi y}f(y)\mathrm{d}y \quad (5.23)$$

for any function f defined on $(0, \infty)$. Applying this to each term of J_2, first separating the factor of $1/(s+\gamma)$ in the second and third terms, yields

$$(\mathscr{A}^{-1}J_2)(u) = v_2(0)\mathrm{e}^{-\gamma u} + \frac{2}{\sigma^2}\int_0^u \mathrm{e}^{-\gamma(u-x)}\mathrm{e}^{\xi x}$$
$$\times \int_x^\infty \mathrm{e}^{-\xi y}[\lambda Q(y) - v_1(y)]\mathrm{d}y\mathrm{d}x, \quad u \geq 0. \quad (5.24)$$

A final computation reduces the term involving Q further:

$$\frac{2\lambda}{\sigma^2}\int_0^u \mathrm{e}^{-\gamma(u-z)}\mathrm{e}^{\xi z}\int_z^\infty \mathrm{e}^{-\xi x}Q(x)\mathrm{d}x\mathrm{d}z$$
$$= \frac{2\lambda}{\sigma^2}\mathrm{e}^{-\gamma u}\int_0^u \mathrm{e}^{(\gamma+\xi)z}\int_z^\infty \mathrm{e}^{-\xi x}\left[\int_x^\infty v_2(x-s)p(s)\mathrm{d}s\right]\mathrm{d}x\mathrm{d}z$$
$$= \frac{2\lambda}{\sigma^2}\mathrm{e}^{-\gamma u}\int_0^\infty v_2(-z)\int_z^{u+z}\mathrm{e}^{-\gamma(z-y)}\mathrm{e}^{\xi y}\left[\int_y^\infty \mathrm{e}^{-\xi s}p(s)\mathrm{d}s\right]\mathrm{d}y\mathrm{d}z$$
$$= \int_0^\infty v_2(-z)\left[g(u+z) - \mathrm{e}^{-\gamma u}g(z)\right]\mathrm{d}z$$
$$= \int_u^\infty v_2(u-z)g(z)\mathrm{d}z - \mathrm{e}^{-\gamma u}\int_0^\infty v_2(-z)g(z)\mathrm{d}z.$$

Using this simplification, substituting Eqs. (5.24) and (5.22) into Eq. (5.21), and finally taking inverse Laplace transform implies Eq. (5.18) and completes the proof. □

5.2. Approximations for phase-type claims

In this subsection, we focus on the expansion terms $\Phi_{0,1}$ when claim sizes are phase-type distributed.

Lemma 5.1 can be applied to the IDEs (5.9) and (5.12) for Φ_0 and Φ_1, provided the integrability requirements $\int_0^\infty \mathrm{e}^{-\xi u}\Phi_0'''(u)\mathrm{d}u < \infty$, $\widetilde{\Phi}_0(\xi) < \infty$ and $\widetilde{\Phi}_1(\xi) < \infty$ are satisfied. Under these assumptions, Eq. (5.21) further implies that the Laplace transforms of Φ_0 and Φ_1 are given by

$$\widetilde{\Phi}_0(s) = \frac{\frac{\sigma^2}{2}\omega(0)(s-\xi) + \lambda(L(s) - L(\xi))}{\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta} \quad \text{and} \quad (5.25)$$

$$\widetilde{\Phi}_1(s) = \frac{\sqrt{2}}{2}\rho v\langle f\phi'\rangle\frac{L_1(s) - L_1(\xi)}{\frac{\sigma^2}{2}s^2 + cs + \lambda(\widetilde{p}(s) - 1) - \delta}, \quad (5.26)$$

where

$$L(s) = -\int_0^\infty \mathrm{e}^{-sx}\int_x^\infty \omega(x-z)p(z)\mathrm{d}z\mathrm{d}x \quad \text{and}$$
$$L_1(s) = s^3\widetilde{\Phi}_0(s) - s^2\omega(0) + s\left(\frac{2\lambda}{\sigma^2}L(\xi) + \gamma\omega(0)\right).$$

When the claim size distribution is phase-type with representation $PH(\boldsymbol{\mu}, \mathbf{A})$ then $\widetilde{p}(s) = \boldsymbol{\mu}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{a}$ and

$$L(s) - L(\xi) = (s-\xi)\boldsymbol{\mu}(s\mathbf{I} - \mathbf{A})^{-1}(\xi\mathbf{I} - \mathbf{A})^{-1}\left[\int_0^\infty \omega(-l)\mathrm{e}^{\mathbf{A}l}\mathrm{d}l\right]\mathbf{a}.$$

Moreover, $\widetilde{\Phi}_0(s)$ and $\widetilde{\Phi}_1(s)$ are rational. In fact, if $\{\rho_i\}_{i=1}^n$ are all roots of the Lundberg fundamental equation (2.12) with negative real part, then $\Phi_0(u)$ and $\Phi_1(u)$ can be written as

$$\Phi_j(u) = \sum_{i=1}^n Q_{ij}(u)\mathrm{e}^{\rho_i u}, \quad u > 0, \quad \text{and} \quad j = 0, 1 \quad (5.27)$$

where $Q_{ij}(u)$ are polynomials. If however, the roots (with negative real parts) are distinct, Chen et al. (2007) point out that the coefficients $\{Q_{i0};\ 1 \leq i \leq n\}$ for $\Phi_0(u)$ are constants and provide an algorithm for calculating them. Interestingly, even in the case of distinct roots, the coefficients $\{Q_{i1};\ 1 \leq i \leq n\}$ for $\Phi_1(u)$ may not be constants. The hyper-exponential example in Section 5.4.1 serves as a prime example of this effect.

Although the representation (5.27) always holds for phase-type claims, it is not always an easy task to find every root with negative real parts for arbitrary sub-intensity matrices **A**. Nonetheless, some specific cases can be computed fully. In Section 5.4, we explore two such examples.

We now turn to the most interesting implication of Lemma 5.1—the representation of the correction term Φ_1 in the EDP function. This is captured in the following Corollary.

Corollary 5.1. *When* $\int_0^\infty e^{-\xi u}\Phi_0'''(u)du < \infty$, $\widetilde{\Phi_0}(\xi) < \infty$ *and* $\widetilde{\Phi_1}(\xi) < \infty$,

$$\Phi_1(u) = \omega_2 q(H * K)(u), \tag{5.28}$$

for $u \geq 0$, *where* $\omega_2 = -\frac{\sqrt{2}\rho v}{\sigma^2 q^2 \gamma}\langle f\phi'\rangle$ *and*

$$H(u) \triangleq \gamma e^{-\gamma u}\int_0^u e^{(\gamma+\xi)z}\int_z^\infty \Phi_0'''(x)e^{-\xi x}\,dxdz.$$

Proof. A direct application of Lemma 5.1 to the IDE for Φ_1 (see Eq. (5.12)) leads to the above result.

For general claim size distributions, $K(u)$ is rather complex. The resulting expressions for $\Phi_0(u)$ and $\Phi_1(u)$ are also complicated and likely impossible to be simplified. Consequently, we focus on phase-type distributed claims from this point forward.

When the claim size distribution is phase-type, $\Phi_0(u) = \Psi(u)$ and the explicit result is provided in Theorem 2.1. We therefore focus on computing $\Phi_1(u)$. Using the notation in Section 2.1, we introduce some additional vectors and matrices:

$$\boldsymbol{\alpha}_2 = [(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})\mathbf{V}^3(\xi\mathbf{I}-\mathbf{V})^{-1}, 0], \qquad \mathbf{T}_2 = \begin{pmatrix}\mathbf{V} & q\mathbf{b}\\ \mathbf{0}^t & -\gamma\end{pmatrix},$$

$$\boldsymbol{\alpha}_3 = [\omega_1\boldsymbol{\alpha}_1\mathbf{T}_1^3(\xi\mathbf{I}-\mathbf{T}_1)^{-1}, 0], \qquad \mathbf{T}_3 = \begin{pmatrix}\mathbf{T}_1 & q\begin{pmatrix}\gamma\\ \mathbf{b}\end{pmatrix}\\ \mathbf{0}^t & -\gamma\end{pmatrix},$$

$$\boldsymbol{\alpha}_{4+i} = [\boldsymbol{\alpha}_{2+i}, \mathbf{0}], \quad \text{and} \quad \mathbf{T}_{4+i} = \begin{pmatrix}\mathbf{T}_{2+i} & \mathbf{t}_{2+i}\boldsymbol{\vartheta}\\ \mathbf{O} & \mathbf{V}\end{pmatrix}, \quad \text{for } i = 0, 1,$$

where $\mathbf{t}_{2+i} = -\mathbf{T}_{2+i}\mathbf{1}$ and $\mathbf{O}$ is a matrix of all 0's. Finally, define functions

$$H_k(u) = \boldsymbol{\alpha}_k\mathbf{1} - \boldsymbol{\alpha}_k e^{\mathbf{T}_k u}\mathbf{1}, \quad u \geq 0 \text{ and } 1 \leq k \leq 5. \tag{5.29}$$

Proposition 5.2. *When the claim size distribution is phase-type with representation PH*$(\boldsymbol{\mu}, \mathbf{A})$, $\Phi_1(u)$ *is given by*

$$\Phi_1(u) = \omega_2[(\boldsymbol{\alpha}_2\mathbf{1} + \boldsymbol{\alpha}_3\mathbf{1})H_1(u) - H_4(u) - H_5(u)], \quad u \geq 0. \tag{5.30}$$

Proof. For any non-singular sub-intensity matrix $\mathbf{E}$ and vector $\boldsymbol{\alpha}_0$,

$$\begin{aligned}&\gamma e^{-\gamma u}\int_0^u e^{(\gamma+\xi)z}\int_z^\infty e^{-\xi x}\boldsymbol{\alpha}_0 e^{\mathbf{E}x}\mathbf{1}\,dxdz\\ &= \int_0^u \gamma e^{-\gamma(u-z)}\boldsymbol{\alpha}_0(\xi\mathbf{I}-\mathbf{E})^{-1}e^{\mathbf{E}z}\mathbf{1}\,dz\\ &= (\boldsymbol{\alpha}_0(\xi\mathbf{I}-\mathbf{E})^{-1}e^{\mathbf{E}u}\mathbf{1}) * (1 - e^{-\gamma u}).\end{aligned}$$

When the claim size distribution is phase-type, Eq. (2.14) implies $\Phi_0'''(u)$ is bounded for $u \geq 0$, and Eq. (5.28) reduces to

$$\begin{aligned}\Phi_1(u) = \omega_2[&(\boldsymbol{\alpha}_2\mathbf{1} + \boldsymbol{\alpha}_3\mathbf{1})(1 - e^{-\gamma u})\\ &- H_2(u) - H_3(u)] * K(u), \quad u \geq 0.\end{aligned}$$

In analogy with Eq. (2.17), $\Phi_1(u)$ can be rewritten as the r.h.s. of Eq. (5.30). □

When the penalty function $\omega(x) = 1$, then $\boldsymbol{\nu}_\omega = \boldsymbol{\vartheta}$, and $\omega_1 = q$. Therefore, $\boldsymbol{\alpha}_2 = \mathbf{0}^t$, then $\boldsymbol{\alpha}_4 = \mathbf{0}^t$, and Eq. (5.30) results in a rather compact representation of the correction to the Laplace transform of ruin time with stochastic volatility:

$$\Phi_1(u) = \omega_2[\boldsymbol{\alpha}_5 e^{\mathbf{T}_5 u}\mathbf{1} - \boldsymbol{\alpha}_5\mathbf{1}\boldsymbol{\alpha}_1 e^{\mathbf{T}_1 u}\mathbf{1}], \quad u \geq 0. \tag{5.31}$$

5.3. Accuracy of the approximation

Thus far, we have obtained the first two terms Φ_0 and Φ_1 of the asymptotic expansion of the EDP function Φ^ϵ under our SV extended risk model. Both terms are found to be independent of the driving source Y_t of volatility $f(Y_t)$. We further demonstrated that when the claim size distribution is phase-type a matrix-type representation exists. The last remaining task is to determine the approximation error $|\Phi^\epsilon(u,y) - (\Phi_0(u) + \sqrt{\epsilon}\,\Phi_1(u))|$ induced by truncating the expansion series at order $\epsilon^{1/2}$. This is addressed in the following Proposition.

Proposition 5.3. *If* $f^2(y)$ *is at most linearly increasing in* $|y|$ *and the claim size distribution is phase-type, then*

$$|\Phi^\epsilon(u,y) - (\Phi_0(u) + \sqrt{\epsilon}\Phi_1(u))| \leq \epsilon(C_1 + C_2|y|^2) \tag{5.32}$$

for some non-negative constants C_1 *and* C_2.

Proof. Define the residual term $Z^\epsilon(u,y)$ by

$$\begin{aligned}Z^\epsilon(u,y) \triangleq\ &\Phi^\epsilon(u,y) - \Phi_0(u) - \epsilon^{1/2}\Phi_1(u)\\ &- \epsilon\Phi_2(u,y) - \epsilon^{3/2}\Phi_3(u,y).\end{aligned}$$

Our first task is to provide a probabilistic representation of this error term.

Due to the boundary condition imposed on $\Phi_0(u)$, $\Phi_1(u)$, $\Phi_2(u,y)$ and $\Phi_3(u,y)$ stated in Section 3, we have

$$Z^\epsilon(u,y) = -\epsilon\left(\Phi_2(0,y) + \sqrt{\epsilon}\,\Phi_3(0,y)\right), \quad \text{for } u \leq 0.$$

For $u > 0$, since $(\mathcal{L}-\delta)\Phi^\epsilon(u,y) = 0$, together with the power matching carried out in Section 5.1, we have

$$(\mathcal{L}-\delta)Z^\epsilon(u,y) = -\epsilon F^\epsilon(u,y)$$

where $F^\epsilon(u,y) \triangleq \mathcal{L}_1\Phi_3(u,y) + \mathcal{L}_2\Phi_2(u,y) + \sqrt{\epsilon}\mathcal{L}_2\Phi_3(u,y)$. As in Propositions 4.1 and 5.1, $Z^\epsilon(u,y)$ enjoys the probabilistic representation

$$\begin{aligned}Z^\epsilon(u,y) = \epsilon\,\mathbb{E}\Bigg[&-e^{-\delta\tau}\left(\Phi_2(0,Y_\tau) + \sqrt{\epsilon}\,\Phi_3(0,Y_\tau)\right)\mathbb{I}(\tau<\infty)\\ &+\int_0^\tau e^{-\delta s}F^\epsilon(U_s,Y_s)ds\,\Big|\,U_0 = u, Y_0 = y\Bigg].\end{aligned} \tag{5.33}$$

Given the above probabilistic representation, we now obtain bounds when the claim distribution is phase-type. Under this assumption, $\Phi_0(u)$ and $\Phi_1(u)$ can be rewritten as in Eq. (5.27). When $f^2(y)$ is at most linearly increasing in $|y|$, so is $\mathcal{L}_2\Phi_0(u)$. Since $\Phi_2(u,y)$ satisfies the Poisson Equation (5.8), the growth property in Eq. (3.12) implies that $\Phi_2(u,y)$ is uniformly bounded in u and at most linearly increasing in $|y|$. Therefore,

$$\begin{aligned}\mathcal{L}_1\Phi_2(u,y) + \mathcal{L}_2\Phi_1(u) =\ &\sqrt{2}\rho v f(y)\frac{\partial^2}{\partial u\partial y}\Phi_2(u,y)\\ &+ (\mathcal{L}_c(f(y)) - \delta)\,\Phi_1(u),\end{aligned}$$

inherits this property.

Furthermore, $\Phi_3(u,y)$ satisfies the Poisson Equation (5.7) together with the growth property of Poisson Equations (see Eq. (3.12)), implies that $\Phi_3(u,y)$ is at mostly linearly increasing in $|y|$ and uniformly bounded in u. Consequently, $F^\epsilon(u,y)$ is at mostly linearly increasing in $|y|^2$ and uniformly bounded in u.

Finally, Eq. (5.33) implies that for $u > 0$

$$\begin{aligned}
&|\Phi^\epsilon(u,y) - (\Phi_0(u) + \sqrt{\epsilon}\Phi_1(u))| \\
&\quad = |Z^\epsilon(u,y) + \epsilon(\Phi_2(u,y) + \sqrt{\epsilon}\Phi_3(u,y))| \\
&\quad \le \epsilon(C_{11} + C_{12}|y|) \\
&\qquad + \epsilon\left(C_{21} + C_{22}\mathbb{E}\left[\int_0^\infty \mathrm{e}^{-\delta s}|Y_s|^2 \mathrm{d}s \,\middle|\, U_0 = u, Y_0 = y\right]\right) \\
&\quad \le \epsilon(C_1 + C_2|y|^2)
\end{aligned}$$

for some constants C_{11}, C_{12}, C_{21}, C_{22}, C_1 and C_2. For $u \le 0$, the boundary conditions imposed on Φ_0 and Φ_1 imply $|\Phi^\epsilon(u,y) - (\Phi_0(u) + \sqrt{\epsilon}\Phi_1(u))| = 0$ and the proof is complete. □

5.4. Examples

In this section, we apply the results of the asymptotic expansion to two special cases of phase-type distributed claims: (i) hyper-exponential claims; and (ii) mixture of Erlang claims.

5.4.1. Hyper-exponential claims

In this subsection, we assume claim size is hyper-exponential distributed with parameters $\{\eta_i; 1 \le i \le m\}$, i.e. the density function is

$$p(x) = \sum_{i=1}^m p_i\eta_i \mathrm{e}^{-\eta_i x}\mathbb{I}(x > 0), \tag{5.34}$$

where $\eta_1 > \eta_2 > \cdots > \eta_m > 0, p_i > 0; 1 \le i \le m$ and $\sum_{i=1}^m p_i = 1$, then the Lundberg fundamental equation is

$$\frac{\sigma^2}{2}s^2 + cs + \lambda\sum_{i=1}^m p_i\frac{\eta_i}{\eta_i + s} - (\lambda + \delta) = 0, \tag{5.35}$$

with domain $\mathbb{R} \setminus \{-\eta_i; 1 \le i \le m\}$.

Denote $l(s) \triangleq \frac{\sigma^2}{2}s^2 + cs + \lambda\sum_{i=1}^m p_i\frac{\eta_i}{\eta_i+s} - (\lambda + \delta)$, then

$$-\infty = \lim_{s\uparrow -\eta_i} l(s) < \lim_{s\downarrow -\eta_i} l(s) = \infty, \quad \lim_{s\to\pm\infty} l(s) = \infty. \tag{5.36}$$

In addition,

$$\begin{aligned}
&l(0) = -\delta < 0, \qquad l'(0) = c - \sum_{i=1}^m \frac{\lambda p_i}{\eta_i} > 0, \\
&l''(s) = \sigma^2 + 2\lambda\sum_{i=1}^m p_i\frac{\eta_i}{(\eta_i + s)^3},
\end{aligned} \tag{5.37}$$

then there exist $m + 2$ single roots of the Lundberg fundamental equation ordered as follows:

$$\begin{aligned}
&-\infty < \rho_1 < -\eta_1 < \rho_2 < -\eta_2 < \cdots \\
&\quad < \rho_m < -\eta_m < \rho_{m+1} < 0 < \xi < \infty.
\end{aligned} \tag{5.38}$$

The roots $\{\rho_i; 1 \le i \le m+1\}\bigcup\{\xi\}$ are functions of the volatility $\sigma = \sqrt{\langle f^2\rangle}$ and discount rate δ. Consequently, it is interesting to investigate their sensitivity on these parameters.

Proposition 5.4. *The roots of Eq. (5.35) have properties*

$$s\frac{\partial s}{\partial \sigma} < 0 \quad \textit{and} \quad s\frac{\partial s}{\partial \delta} > 0. \tag{5.39}$$

Proof. Taking derivatives of Eq. (5.35) with respect to $D = \frac{\sigma^2}{2}$:

$$\frac{\partial s}{\partial D} = \frac{s^2}{\lambda\sum_{i=1}^m \frac{p_i\eta_i}{(\eta_i+s)^2} - c - 2sD}.$$

Replacing $Ds^2 + cs$ by using Eq. (5.35), we find

$$\frac{\partial s}{\partial D} = -\frac{s^3}{\lambda s^2\sum_{i=1}^m \frac{p_i}{(\eta_i+s)^2} + s^2 D + \delta}.$$

Since D is increasing in σ, then $s\frac{\partial s}{\partial \sigma} < 0$.

For the sensitivity with respect to δ we use similar analysis to find:

$$\frac{\partial s}{\partial \delta} = \frac{1}{2sD + c - \lambda\sum_{i=1}^m \frac{p_i\eta_i}{(\eta_i+s)^2}} = \frac{s}{\lambda s^2\sum_{i=1}^m \frac{p_i}{(\eta_i+s)^2} + s^2 D + \delta}$$

and the proof is complete. □

Remark 5.1. As the volatility increases, the negative roots are non-decreasing and the positive roots are non-increasing. While the opposite situation arises as the discount rate δ increases.

Since the roots are distinct, Φ_0 and Φ_1 have the form in Eq. (5.27) with $n = m+1$. When $m = 1$, the claim size distribution is exponential, and this case can be computed even more explicitly. To this end, denote $J \triangleq \int_0^\infty \omega(-l)\mathrm{e}^{-\eta_1 l}\mathrm{d}l$, then

$$L(s) = -\frac{\eta_1}{\eta_1 + s}\int_0^\infty \omega(-l)\mathrm{e}^{-\eta_1 l}\mathrm{d}l = -\frac{\eta_1 J}{\eta_1 + s}. \tag{5.40}$$

Eq. (5.25) implies

$$\begin{aligned}
\widetilde{\Phi}_0(s) &= \frac{\frac{\sigma^2}{2}\omega(0)s + \left(\frac{\sigma^2}{2}\omega(0)\eta_1 + \lambda J\eta_1/(\eta_1 + \xi)\right)}{\frac{\sigma^2}{2}(s - \rho_1)(s - \rho_2)} \\
&= \frac{Q^*}{s - \rho_1} + \frac{\omega(0) - Q^*}{s - \rho_2},
\end{aligned} \tag{5.41}$$

where $Q^* = \frac{\omega(0)(\rho_1+\eta_1) + \left(\lambda J\frac{2\eta_1}{\sigma^2(\eta_1+\xi)}\right)}{(\rho_1 - \rho_2)} \le \omega(0)$. Therefore,

$$\Phi_0(u) = Q^*\mathrm{e}^{\rho_1 u} + (\omega(0) - Q^*)\mathrm{e}^{\rho_2 u}, \quad u \ge 0,$$

and so $\Phi_0'''(u) = Q^*\rho_1^3\mathrm{e}^{\rho_1 u} + (\omega(0) - Q^*)\rho_2^3\mathrm{e}^{\rho_2 u}$.

Now using Eq. (5.25), through straight forward calculations we find

$$\begin{aligned}
\widetilde{\Phi}_1(s) &= -\frac{\sqrt{2}}{2}\rho v\langle f\phi'\rangle\left[\frac{Q^*\rho_1^3(s + \eta_1)/(\xi - \rho_1)}{\frac{\sigma^2}{2}(s - \rho_1)^2(s - \rho_2)}\right. \\
&\qquad \left. + \frac{(s + \eta_1)(\omega(0) - Q^*)\rho_2^3/(\xi - \rho_2)}{\frac{\sigma^2}{2}(s - \rho_1)(s - \rho_2)^2}\right] \\
&= -\frac{\sqrt{2}}{\sigma^2}\rho v\langle f\phi'\rangle\left[\frac{Q^*\rho_1^3(\rho_1 + \eta_1)}{(\xi - \rho_1)(\rho_1 - \rho_2)}\frac{1}{(s - \rho_1)^2}\right. \\
&\qquad + \frac{(\omega(0) - Q^*)\rho_2^3(\rho_2 + \eta_1)}{(\xi - \rho_2)(\rho_2 - \rho_1)}\frac{1}{(s - \rho_2)^2} \\
&\qquad + \left\{-\frac{Q^*\rho_1^3(\rho_2 + \eta_1)}{(\xi - \rho_1)(\rho_1 - \rho_2)^2} + \frac{(\omega(0) - Q^*)\rho_2^3(\rho_1 + \eta_1)}{(\xi - \rho_2)(\rho_2 - \rho_1)^2}\right\} \\
&\qquad \left.\times\left(\frac{1}{s - \rho_1} - \frac{1}{s - \rho_2}\right)\right],
\end{aligned}$$

hence, for $u \ge 0$,

$$\begin{aligned}
\Phi_1(u) &= -\frac{\sqrt{2}}{\sigma^2}\rho v\langle f\phi'\rangle\left[\frac{Q^*\rho_1^3(\rho_1 + \eta_1)}{(\xi - \rho_1)(\rho_1 - \rho_2)}u\mathrm{e}^{\rho_1 u}\right. \\
&\qquad \left. + \frac{(\omega(0) - Q^*)\rho_2^3(\rho_2 + \eta_1)}{(\xi - \rho_2)(\rho_2 - \rho_1)}u\mathrm{e}^{\rho_2 u}\right.
\end{aligned}$$

$$+\left\{-\frac{Q^*\rho_1^3(\rho_2+\eta_1)}{(\xi-\rho_1)(\rho_1-\rho_2)^2}+\frac{(\omega(0)-Q^*)\rho_2^3(\rho_1+\eta_1)}{(\xi-\rho_2)(\rho_2-\rho_1)^2}\right\}\times\left(e^{\rho_1 u}-e^{\rho_2 u}\right)\Bigg].$$

5.4.2. Mixed-Erlang claims

For any given positive distribution function $F(x)$ defined on $(0,\infty)$, let

$$\hat{f}(x|\eta)=\sum_{i=1}^{\infty}(F(i/\eta)-F((i-1)/\eta))\tau_i(x,\eta),\quad x>0, \tag{5.42}$$

where $\tau_i(x,\eta)$ is the pdf of Erlang(i, η) and is given by

$$\tau_i(x,\eta)=\frac{\eta(\eta x)^{i-1}e^{-\eta x}}{(i-1)!},\quad x>0, i\geq 1. \tag{5.43}$$

Tijms (1994) shows that $\lim_{\eta\to\infty}\hat{F}(x|\eta)=F(x)$ pointwise for all continuous points x of $F(x)$, where $\hat{F}(x|\eta)$ is the distribution function of $\hat{f}(x|\eta)$.

In this subsection, the claim distribution is assumed to be a mixture of Erlangs, i.e.

$$p(x)=\sum_{i=1}^{\infty}q_i\tau_i(x,\eta),\quad x>0, \tag{5.44}$$

where $q_i\geq 0$ and $\sum_{i=1}^{\infty}q_i=1$. We follow the results in Willmot and Woo (2007) to give expressions of $\Phi_0'(u)$ and $\Phi_1'(u)$.

Proposition 5.5. *When the claim size distribution is mixed-Erlang defined in Eq. (5.44), then there exist constants* $\{q_i^-,q_i^+,q_i^*,q_i^{**}\}_{i=1}^{\infty}$ *and* q^{++} *such that for* $u\geq 0$

$$\Phi_0'(u)=\begin{cases}\omega(0)\delta_0(u)+\sum_{i=1}^{\infty}q_i^-\tau_i(u,\gamma), & \gamma>\eta\\ \omega(0)\delta_0(u)-q^{++}\tau_1(u,\gamma)+\sum_{i=1}^{\infty}q_i^+\tau_i(u,\eta), & \gamma\leq\eta\end{cases} \tag{5.45}$$

and

$$\Phi_1'(u)=\begin{cases}-\frac{\sqrt{2}\rho v\langle f\phi'\rangle}{\sigma^2\gamma}\sum_{i=1}^{\infty}q_i^*\tau_i(u,\gamma), & \gamma>\eta\\ -\frac{\sqrt{2}\rho v\langle f\phi'\rangle}{\sigma^2\gamma}\sum_{i=1}^{\infty}q_i^{**}\tau_i(u,\eta), & \gamma\leq\eta.\end{cases} \tag{5.46}$$

The proof of the above Proposition and the method for computing the coefficients $\{q_i^-,q_i^+,q_i^*,q_i^{**}\}_{i=1}^{\infty}$ and q^{++} are provided in Appendix B.

6. Concluding remarks

In this paper, we consider the EDP function under a new stochastic volatility extended risk model. In this new model, the surplus investment's volatility is driven by an underlying Ornstein–Uhlenbeck process. Although the exact expression for the EDP function seems unattainable, through singular perturbation theory techniques we succeeded in obtaining an asymptotic expansion of the EDP function in powers of inverse rate of mean-reversion. Fast mean-reverting volatility models have been successful in explaining features in both historical prices and implied volatility surfaces for equity derivatives, interest rate derivatives and commodity derivatives. Consequently, we believe that such a model is valid for the surplus process as well.

Although the analysis in Section 5, and particularly Proposition 5.3 fully specifies the truncation error, it would be interesting to investigate the accuracy of the approximation under specific parameters. Numerical solutions and approximations of the EDP function under various insurance risk models with constant volatility have been considered by several authors. A common approach is to use techniques developed in numerical analysis for Volterra integro-differential equations to solve corresponding IDEs similar to Eq. (2.2) (see Albrecher et al. (2003), Albrecher and Kainhofer (2002), Makroglou et al. (1999), Makroglou (2004), and references therein). Another typical approach is to convert the IDEs into renewal equations and solve the renewal equations as integral equations. For example, a functional approximation for the solution of a renewal equation is proposed and demonstrated in Pitts and Politis (2007) and Pitts and Politis (2008). However, numerically solving the nonlinear partial integro-differential equation (4.1) will be much more challenging as the afore-mentioned techniques for IDEs are no longer applicable. As a result, it will require a thorough numerical analysis in its own right. We intend to explore numerical solutions of Eq. (4.1) in our future work.

The proposed new risk model might also be useful in valuing derivatives exposed to credit risk. Specifically, one can assume that the firm value follows the process e^{U_t} where U_t is defined in Eq. (3.1), and default of the firm is triggered when its value drops below a critical level. It is often assumed in credit risk research that the default time and recovery rate are independent. With the approaches developed in this paper, one might be able to analyze the relationship between default time and recovery rate via the EDP function, and to price defaultable bonds and other credit-linked instruments accordingly.

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Appendix A. Useful lemmas

The following two Lemmas, extracted from Ren (2005) and Neuts (1981) and stated without proof, demonstrate that phase-type distributions are closed under convolution.

Lemma A.1. *Let* F_1 *and* F_2 *be phase-type distribution functions with representations* $PH(\boldsymbol{\mu}_1,\mathbf{E}_1)$ *and* $PH(\boldsymbol{\mu}_2,\mathbf{E}_2)$ *respectively, and* $F=F_1*F_2$.

1. *If* F_1 *and* F_2 *both aren't zero-modified, then* F *is a phase-type distribution function with representation* $PH(\boldsymbol{\mu},\mathbf{E})$, *where*

$$\boldsymbol{\mu}=(\|\boldsymbol{\mu}_2\|\boldsymbol{\mu}_1,\mathbf{0}^t),\quad \text{and}\quad \mathbf{E}=\begin{pmatrix}\mathbf{E}_1 & (-\mathbf{E}_1\mathbf{1})\dfrac{\boldsymbol{\mu}_2}{\|\boldsymbol{\mu}_2\|}\\ & \mathbf{E}_2\end{pmatrix}, \tag{A.1}$$

   where $\mathbf{0}$ *is the column vector of zeros.*
2. *If* F_1 *isn't zero-modified and* F_2 *is, then* F *is a phase-type distribution function with representation* $PH(\boldsymbol{\mu},\mathbf{E})$, *where*

$$\boldsymbol{\mu}=(\boldsymbol{\mu}_1,\mathbf{0}^t),\quad \text{and}\quad \mathbf{E}=\begin{pmatrix}\mathbf{E}_1 & (-\mathbf{E}_1\mathbf{1})\boldsymbol{\mu}_2\\ & \mathbf{E}_2\end{pmatrix}. \tag{A.2}$$

Lemma A.2. *Let* F_1 *be a phase-type distribution function with representation* $PH(\boldsymbol{\mu}_1,\mathbf{E})$ *and* F_2 *be zero-modified phase-type*

distribution function with representation $PH(\boldsymbol{\mu}_2, \mathbf{E} + (-\mathbf{E1})\boldsymbol{\mu}_2)$, *then* $F = F_1 * F_2$ *is a phase-type distribution function with representation* $PH(\boldsymbol{\mu}_1, \mathbf{E} + (-\mathbf{E1})\boldsymbol{\mu}_2)$.

The following Lemma is useful in constructing Φ_0 and Φ_1 which approximate the EDP function in the SV extended risk model.

Lemma A.3. *The positive root* ξ *of the Lundberg fundamental equation (2.4) is single.*

Proof. If the positive root is multiple, take derivatives of both sides of Eq. (2.4) w.r.t. s and then take $s = \xi$. Thus,

$$\sigma^2\xi + c - \lambda \int_0^\infty x\mathrm{e}^{-\xi x} p(x)\mathrm{d}x = 0.$$

Multiply both sides of the above equation by ξ and subtract Eq. (2.4), then

$$\begin{aligned} 0 &= \frac{\sigma^2}{2}\xi^2 + \delta + \lambda \int_0^\infty \left[1 - (1+\xi x)\mathrm{e}^{-\xi x}\right] p(x)\mathrm{d}x \\ &> \frac{\sigma^2}{2}\xi^2 + \delta > 0, \end{aligned} \tag{A.3}$$

where the first inequality follows because $1 - (1+\xi x)\mathrm{e}^{-\xi x} \geq 0$ for $x \geq 0$. The proof is completed. □

Appendix B. The mixed-Erlang claim example

The proof of Proposition 5.5 is rather lengthy, we divide it into several Lemmas and Propositions. Meanwhile, the way to calculate the coefficients $\{q_i^-, q_i^+, q_i^*, q_i^{**}\}_{i=1}^\infty$ and q^{++} is given.

Since the claim is mixed-Erlang distributed, Lin and Willmot (1999) show that

$$g_2(x) = \frac{2\lambda}{\sigma^2\gamma} \int_x^\infty \mathrm{e}^{-\xi(y-x)} p(y)\mathrm{d}y = \sum_{i=1}^\infty q_i^{(0)} \tau_i(x, \eta), \tag{B.1}$$

where $q_i^{(0)} = \frac{2\lambda}{\sigma^2\gamma(\eta+\xi)} \sum_{k=0}^\infty q_{k+i} (\frac{\eta}{\eta+\xi})^k$ for $i \geq 1$.

Lemma B.1. *Under the assumptions above, for* $x > 0$

$$g(x) = \begin{cases} \sum_{i=2}^\infty q_i^{(11)} \tau_i(x, \gamma), & \gamma > \eta \\ \sum_{i=2}^\infty q_i^{(10)} \tau_i(x, \eta), & \gamma \leq \eta \end{cases} \tag{B.2}$$

where $q_i^{(10)} = \frac{\gamma}{\eta} \sum_{j=1}^{i-1} q_{i-j}^{(0)} \left(1 - \frac{\gamma}{\eta}\right)^{j-1}$, $q_i^{(11)} = \sum_{j=1}^{i-1} \binom{i-2}{j-1} \left(\frac{\eta}{\gamma}\right)^j \left(1 - \frac{\eta}{\gamma}\right)^{i-j-1} q_j^{(0)}$, $i \geq 2$.

Proof. 1. When $\gamma \leq \eta$, Willmot and Woo (2007) show that

$$\widetilde{g}_1(s) = \frac{\gamma}{s+\gamma} = \frac{\eta}{\eta+s} \left\{ \frac{\gamma/\eta}{1 - (1 - \gamma/\eta)\frac{\eta}{\eta+s}} \right\},$$

then

$$\begin{aligned} \widetilde{g}(s) = \widetilde{g}_1(s)\widetilde{g}_2(s) &= \frac{\eta}{\eta+s} \left\{ \frac{\gamma/\eta}{1 - (1 - \gamma/\eta)\frac{\eta}{\eta+s}} \right\} \\ &\quad \times \left\{ \sum_{i=1}^\infty q_i^{(0)} \left(\frac{\eta}{\eta+s}\right)^i \right\} = \sum_{i=2}^\infty q_i^{(10)} \left(\frac{\eta}{\eta+s}\right)^i. \end{aligned}$$

Thus, inversion Laplace transform implies Eq. (B.2).

2. When $\gamma > \eta$,

$$\frac{\eta}{s+\eta} = \frac{\gamma}{\gamma+s} \left\{ \frac{\eta/\gamma}{1 - (1 - \eta/\gamma)\frac{\gamma}{\gamma+s}} \right\}.$$

For notational convenience, for $t \in (0, 1]$, negative binomial type probability $h_m(i, t)$ is defined in Willmot and Woo (2007) as

$$h_m(i, t) = \binom{i+m-1}{m} t^i (1-t)^m, \quad m = 0, 1, 2, \ldots.$$

The fact $\sum_{m=0}^\infty h_m(i, t) z^m = \left\{ \frac{t}{1-(1-t)z} \right\}^i$ implies that

$$\begin{aligned} \widetilde{g}(s) &= \sum_{i=1}^\infty q_i^{(0)} \left(\frac{\gamma}{\gamma+s}\right)^{i+1} \left\{ \frac{\eta/\gamma}{1 - (1 - \eta/\gamma)\frac{\gamma}{\gamma+s}} \right\}^i \\ &= \sum_{i=1}^\infty q_i^{(0)} \sum_{m=0}^\infty h_m\left(i, \frac{\eta}{\gamma}\right) \left(\frac{\gamma}{\gamma+s}\right)^{i+1+m}. \end{aligned}$$

Thus, $\widetilde{g}(s) = \sum_{i=2}^\infty q_i^{(11)} \left(\frac{\gamma}{\gamma+s}\right)^i$ according to the definition of $\{q_i^{(11)}, i \geq 2\}$, then we obtain Eq. (B.2) via the inversion of the Laplace transform. □

Lemma B.2. *Under the assumptions above, for* $x \geq 0$

$$K'(x) = \begin{cases} q\delta_0(x) + \sum_{i=2}^\infty q_i^{(21)} \tau_i(x, \gamma), & \gamma > \eta \\ q\delta_0(x) + \sum_{i=2}^\infty q_i^{(20)} \tau_i(x, \eta), & \gamma \leq \eta \end{cases} \tag{B.3}$$

where

$$\begin{aligned} &q_0^{(2k)} = q, \qquad q_1^{(2k)} = 0, \\ &q_i^{(2k)} = \sum_{j=2}^i q_{i-j}^{(2k)} q_j^{(1k)}, \quad i = 2, 3, \ldots, k = 0, 1. \end{aligned} \tag{B.4}$$

Proof. When $\gamma \leq \eta$, $\sum_{i=2}^\infty q_i^{(10)}/(1-q) = G(\infty)/(1-q) = 1$ and

$$\begin{aligned} \int_{0-}^\infty \mathrm{e}^{-su}\mathrm{d}K(u) &= \frac{q}{1 - \widetilde{g}(s)} \\ &= \left\{ 1 - \frac{1-q}{q} \left(\sum_{i=2}^\infty \frac{q_i^{(10)}}{1-q} \left\{ \frac{\eta}{\eta+s} \right\}^i - 1 \right) \right\}^{-1} \\ &= \sum_{i=0}^\infty d_i \left\{ \frac{\eta}{\eta+s} \right\}^i \end{aligned}$$

where d_i is given by equation

$$\sum_{i=0}^\infty d_i z^i = \left\{ 1 - \frac{1-q}{q} \left(\sum_{i=2}^\infty \frac{q_i^{(10)}}{1-q} z^i - 1 \right) \right\}^{-1}, \quad \forall z \in (0, 1). \tag{B.5}$$

From the above equation, we find that $\{d_i; i \geq 0\}$ is the p.d.f. of a compound geometric distribution. Theorem 4.48 in Klugman et al. (2004) shows $d_i = q_i^{(20)}$ defined in Eq. (B.4) and provides a recursive algorithm. We complete the proof for $\gamma > \eta$ case via the same analysis. □

Proposition B.1. *Denote the forward difference operator as* $\Delta d_i = d_{i+1} - d_i$. *Under the assumptions above, let* $q_1^{(1k)} = 0$, $k = 0, 1$ *and*

denote

$$h_j = \sum_{k=1}^{\infty} q_{j+k-1}^{(10)} \int_0^{\infty} \tau_k(y,\eta)\omega(-y)\mathrm{d}y, \quad j = 1, 2, \ldots$$

and

$$h_j' = \sum_{k=1}^{\infty} q_{j+k-1}^{(11)} \int_0^{\infty} \tau_k(y,\gamma)\omega(-y)\mathrm{d}y, \quad j = 1, 2, \ldots,$$

then for $u \geq 0$

$$\Phi_0'(u) = \begin{cases} \omega(0)\delta_0(u) + \sum_{i=1}^{\infty} q_i^{(31)} \tau_i(u,\gamma), & \gamma > \eta \\ \omega(0)\delta_0(u) - (\omega(0) - h_1)\gamma \mathrm{e}^{-\gamma u} & \\ \quad + \sum_{i=1}^{\infty} q_i^{(30)} \tau_i(u,\eta), & \gamma \leq \eta \end{cases} \tag{B.6}$$

where $q_1^{(30)} = \Delta h_1$, $q_2^{(30)} = \Delta h_2 + \omega(0) q_2^{(20)}/q$ *and*

$$q_i^{(30)} = \Delta h_i + \omega(0) q_i^{(20)}/q + \sum_{j=1}^{i-2} \left\{ \Delta h_j - (\omega(0) - h_1) \times \frac{\gamma}{\eta}\left(1 - \frac{\gamma}{\eta}\right)^{j-1} \right\} q_{i-j}^{(20)}/q, \quad i = 3, 4, \ldots,$$

and $q_1^{(31)} = h_2' - \omega(0)$, $q_2^{(31)} = \Delta h_2' + \omega(0) q_2^{(21)}/q$ *and*

$$q_i^{(31)} = \Delta h_i' + \omega(0) q_i^{(21)}/q - (\omega(0) - h_1') q_{i-1}^{(21)}/q + \sum_{j=1}^{i-2} \Delta h_j' q_{i-j}^{(21)}/q, \quad i = 3, 4, \ldots.$$

Proof. 1. When $\gamma \leq \eta$, Willmot and Woo (2007) point out

$$g(x+y) = \frac{1}{\eta} \sum_{j=1}^{\infty} \left\{ \sum_{k=1}^{\infty} q_{j+k-1}^{(10)} \tau_k(x,\eta) \right\} \tau_j(y,\eta), \tag{B.7}$$

then

$$\int_u^{\infty} \omega(u-y) g(y)\mathrm{d}y = \int_0^{\infty} \omega(-y) g(u+y)\mathrm{d}y = \frac{1}{\eta} \sum_{j=1}^{\infty} \left\{ \sum_{k=1}^{\infty} q_{j+k-1}^{(10)} \int_0^{\infty} \tau_k(y,\eta)\omega(-y)\mathrm{d}y \right\} \tau_j(u,\eta).$$

It is easy to verify that $\int_0^{\infty} \omega(-y) g(y)\mathrm{d}y = h_1$ and

$$\frac{1}{\eta} \sum_{j=1}^{\infty} h_j \tau_j(u,\eta) - h_1 = \int_0^u \sum_{j=1}^{\infty} \Delta h_j \tau_j(y,\eta)\mathrm{d}y.$$

The proof of Lemma B.1 shows that

$$\frac{\gamma}{s+\gamma} \left\{ \sum_{i=2}^{\infty} q_i^{(20)} \left(\frac{\eta}{s+\eta}\right)^i \right\} = \sum_{i=3}^{\infty} \left\{ \sum_{j=1}^{i-2} \frac{\gamma}{\eta}\left(1 - \frac{\gamma}{\eta}\right)^{j-1} q_{i-j}^{(20)} \right\} \left(\frac{\eta}{s+\eta}\right)^i.$$

According to Eq. (2.8), Lemma B.2 implies

$$\int_{0-}^{\infty} \mathrm{e}^{-su}\mathrm{d}\Psi(u) = \left[\omega(0) - (\omega(0) - h_1)\frac{\gamma}{\gamma+s} + \sum_{j=1}^{\infty} \Delta h_j \left(\frac{\eta}{s+\eta}\right)^j \right] \times \left[1 + \sum_{i=2}^{\infty} \frac{q_i^{(20)}}{q} \left(\frac{\eta}{s+\eta}\right)^i \right]$$
$$= \omega(0) - (\omega(0) - h_1)\frac{\gamma}{\gamma+s} + \sum_{i=1}^{\infty} q_i^{(30)} \left(\frac{\eta}{s+\eta}\right)^i.$$

The fact $\Phi_0(u) = \Psi(u)$ implies Eq. (B.6).

2. When $\gamma > \eta$, using similar analysis, we get $\int_0^{\infty} \omega(-y) g(y)\mathrm{d}y = h_1'$ and

$$\int_u^{\infty} \omega(u-y) g(y)\mathrm{d}y - \int_0^{\infty} \omega(-y) g(y)\mathrm{d}y = \int_0^u \sum_{j=1}^{\infty} \Delta h_j' \tau_j(y,\gamma)\mathrm{d}y. \tag{B.8}$$

It is easy to see

$$\frac{\gamma}{s+\gamma} \left\{ \sum_{i=2}^{\infty} q_i^{(21)} \left(\frac{\gamma}{s+\gamma}\right)^i \right\} = \sum_{i=3}^{\infty} q_{i-1}^{(21)} \left(\frac{\gamma}{s+\gamma}\right)^i. \tag{B.9}$$

According to Eq. (2.8), Lemma B.2 implies

$$\int_{0-}^{\infty} \mathrm{e}^{-su}\mathrm{d}\Psi(u) = \left[\omega(0) - (\omega(0) - h_1')\frac{\gamma}{\gamma+s} + \sum_{j=1}^{\infty} \Delta h_j' \left(\frac{\gamma}{s+\gamma}\right)^j \right] \times \left[1 + \sum_{i=2}^{\infty} \frac{q_i^{(21)}}{q} \left(\frac{\gamma}{s+\gamma}\right)^i \right]$$
$$= \omega(0) + \sum_{i=1}^{\infty} q_i^{(31)} \left(\frac{\gamma}{s+\gamma}\right)^i.$$

The proof is complete. □

Remark B.1. The above Proposition provides the way to calculate the coefficients $\{q_i^-, q_i^+\}_{i=1}^{\infty}$ and q^{++} in Proposition 5.5 by

$$q^{++} = \omega(0) - h_1, \qquad q_i^- = q_i^{(31)} \quad \text{and} \quad q_i^+ = q_i^{(30)}, \quad \forall i \geq 1.$$

Proposition B.2. *Under the assumptions above, denote*

$$Q_i = \eta^3 \sum_{k=0}^{\infty} \Delta^2 q_{k+i}^{(30)} \frac{1}{\eta+\xi} \left(\frac{\eta}{\eta+\xi}\right)^k, \quad i = 1, 2, \ldots,$$

and

$$Q_i' = \gamma^3 \sum_{k=0}^{\infty} \Delta^2 q_{k+i}^{(31)} \frac{1}{\gamma+\xi} \left(\frac{\gamma}{\gamma+\xi}\right)^k, \quad i = 1, 2, \ldots,$$

then

$$\Phi_1'(u) = \begin{cases} \dfrac{\sqrt{2}\rho v \langle f\phi' \rangle}{\sigma^2\gamma} \sum_{i=1}^{\infty} q_i^{(41)} \tau_i(u,\gamma), & \gamma > \eta \\ -\dfrac{\sqrt{2}\rho v \langle f\phi' \rangle}{\sigma^2\gamma} \sum_{i=1}^{\infty} q_i^{(40)} \tau_i(u,\eta), & \gamma \leq \eta \end{cases} \tag{B.10}$$

where $q_1^{(40)} = \frac{\gamma^4(h_1 - \omega(0))}{\eta(\gamma+\xi)} + \frac{Q_1\gamma}{\eta}$ *and*

$$q_i^{(40)} = \sum_{j=1}^{i-1} \frac{\gamma}{\eta}\left(1 - \frac{\gamma}{\eta}\right)^{i-j-1} \left\{ \sum_{k=1}^{j-1} \Delta Q_k q_{j-k}^{(20)}/q + \Delta Q_j + Q_1 q_j^{(20)}/q \right\} + Q_1 \frac{\gamma}{\eta}\left(1 - \frac{\gamma}{\eta}\right)^{i-1} + \frac{\gamma^3(h_1 - \omega(0))}{\gamma+\xi} \times \left\{ \sum_{j=2}^{i-1} \frac{\gamma}{\eta}\left(1 - \frac{j\gamma}{\eta}\right)\left(1 - \frac{\gamma}{\eta}\right)^{j-2} q_{i-j}^{(20)}/q + \frac{\gamma}{\eta}\left(1 - \frac{i\gamma}{\eta}\right)\left(1 - \frac{\gamma}{\eta}\right)^{i-2} + \frac{\gamma}{\eta} q_{i-1}^{(20)}/q \right\}, \quad i \geq 2$$

and $q_1^{(41)} = Q_1'$, $q_2^{(41)} = \Delta Q_1' + Q_1' q_1^{(21)}/q$ *and*

$$q_i^{(41)} = \Delta Q_{i-1}' + \sum_{j=1}^{i-2} \Delta Q_{i-j-1}' q_j^{(21)}/q + Q_1' q_{i-1}^{(21)}/q, \quad i = 3, 4, \ldots.$$

Proof. 1. When $\gamma > \eta$, then

$$\Phi_0'''(u) = \gamma^2 \sum_{i=1}^{\infty} \Delta^2 q_i^{(31)} \tau_i(u, \gamma), \quad u > 0. \tag{B.11}$$

Similar to Eq. (B.1), we get

$$\begin{aligned} e^{\xi z} \int_z^{\infty} \Phi_0'''(x) e^{-\xi x} dx &= \gamma^2 \sum_{i=1}^{\infty} \left\{ \sum_{k=0}^{\infty} \Delta^2 q_{k+i}^{(31)} \right. \\ &\quad \left. \times \frac{1}{\gamma+\xi} \left(\frac{\gamma}{\gamma+\xi} \right)^k \right\} \tau_i(z, \gamma) = \sum_{i=1}^{\infty} \frac{Q_i'}{\gamma} \tau_i(z, \gamma), \end{aligned} \tag{B.12}$$

then $\int_0^{\infty} \Phi_0'''(x) e^{-\xi x} dx = Q_1'$ and

$$\begin{aligned} &e^{\xi z} \int_z^{\infty} \Phi_0'''(x) e^{-\xi x} dx - \int_0^{\infty} \Phi_0'''(x) e^{-\xi x} dx \\ &= \int_0^z \sum_{i=1}^{\infty} \Delta Q_i' \tau_i(s, \gamma) ds. \end{aligned}$$

Thus, the Laplace transform of $\Phi_1(u)$ is

$$\begin{aligned} \int_{0-}^{\infty} e^{-su} d\Phi_1(u) &= -\frac{\sqrt{2}\rho v \langle f \phi' \rangle}{\sigma^2 \gamma} \frac{\gamma}{\gamma+s} \\ &\quad \times \left\{ Q_1' + \sum_{i=1}^{\infty} \Delta Q_i' \left(\frac{\gamma}{\gamma+s} \right)^i \right\} \\ &\quad \times \left\{ 1 + \sum_{i=1}^{\infty} q_i^{(21)}/q \left(\frac{\gamma}{\gamma+s} \right)^i \right\} \\ &= -\frac{\sqrt{2}\rho v \langle f \phi' \rangle}{\sigma^2 \gamma} \sum_{i=1}^{\infty} q_i^{(41)} \left(\frac{\gamma}{\gamma+s} \right)^i, \end{aligned}$$

then the inversion of the Laplace transform results in Eq. (B.10);

2. When $\gamma \le \eta$, then

$$\begin{aligned} \Phi_0'''(u) &= (h_1 - \omega(0)) \gamma^3 e^{-\gamma u} \\ &\quad + \eta^2 \sum_{i=1}^{\infty} \Delta^2 q_i^{(30)} \tau_i(u, \eta), \quad u > 0. \end{aligned} \tag{B.13}$$

Similar to Eq. (B.1), we get

$$\begin{aligned} e^{\xi z} \int_z^{\infty} \Phi_0'''(x) e^{-\xi x} dx &= \frac{\gamma^3 (h_1 - \omega(0))}{\gamma+\xi} e^{-\gamma z} \\ &\quad + \eta^2 \sum_{i=1}^{\infty} \left\{ \sum_{k=0}^{\infty} \Delta^2 q_{k+i}^{(30)} \frac{1}{\eta+\xi} \left(\frac{\eta}{\eta+\xi} \right)^k \right\} \tau_i(z, \eta) \\ &= \frac{\gamma^3 (h_1 - \omega(0))}{\gamma+\xi} e^{-\gamma z} + \sum_{i=1}^{\infty} \frac{Q_i}{\eta} \tau_i(z, \eta). \end{aligned}$$

Similar analysis for case $\gamma > \eta$, we get the Laplace transform of $\Phi_1(u)$ as

$$\int_{0-}^{\infty} e^{-su} d\Phi_1(u) = -\frac{\sqrt{2}\rho v \langle f \phi' \rangle}{\sigma^2 \gamma} (I + II), \tag{B.14}$$

where

$$\begin{aligned} I &= \frac{\gamma}{\gamma+s} \left\{ Q_1 + \sum_{i=1}^{\infty} \Delta Q_i \left(\frac{\eta}{\eta+s} \right)^i \right\} \\ &\quad \times \left\{ 1 + \sum_{i=1}^{\infty} q_i^{(20)}/q \left(\frac{\eta}{\eta+s} \right)^i \right\} = \frac{Q_1 \gamma}{\eta} \frac{\eta}{\eta+s} \\ &\quad + \sum_{i=2}^{\infty} \left\{ \sum_{j=1}^{i-1} \frac{\gamma}{\eta} \left(1 - \frac{\gamma}{\eta} \right)^{i-j-1} \left\{ \sum_{k=1}^{j-1} \Delta Q_k q_{j-k}^{(20)}/q \right. \right. \\ &\quad \left. \left. + \Delta Q_j + Q_1 q_j^{(20)}/q \right\} + Q_1 \frac{\gamma}{\eta} \left(1 - \frac{\gamma}{\eta} \right)^{i-1} \right\} \left(\frac{\eta}{\eta+s} \right)^i, \end{aligned}$$

and

$$\begin{aligned} II &= \frac{\gamma^3 (h_1 - \omega(0))}{\gamma+\xi} \left\{ \frac{\gamma}{\gamma+s} - \frac{\gamma^2}{(\gamma+s)^2} \right\} \\ &\quad \times \left\{ 1 + \sum_{i=1}^{\infty} \frac{q_i^{(20)}}{q} \left(\frac{\eta}{\eta+s} \right)^i \right\} \\ &= \frac{\gamma^3 (h_1 - \omega(0))}{\gamma+\xi} \left\{ \frac{\gamma}{\eta} \frac{\eta}{\eta+s} + \sum_{i=2}^{\infty} \frac{\gamma}{\eta} \left(1 - \frac{i\gamma}{\eta} \right) \right. \\ &\quad \left. \times \left(1 - \frac{\gamma}{\eta} \right)^{i-2} \left(\frac{\eta}{\eta+s} \right)^i \right\} \left\{ 1 + \sum_{i=1}^{\infty} \frac{q_i^{(20)}}{q} \left(\frac{\eta}{\eta+s} \right)^i \right\} \\ &= \frac{\gamma^3 (h_1 - \omega(0))}{\gamma+\xi} \left\{ \frac{\gamma}{\eta} \frac{\eta}{\eta+s} + \sum_{i=2}^{\infty} \left\{ \sum_{j=2}^{i-1} \frac{\gamma}{\eta} \left(1 - \frac{j\gamma}{\eta} \right) \right. \right. \\ &\quad \times \left(1 - \frac{\gamma}{\eta} \right)^{j-2} q_{i-j}^{(20)}/q + \frac{\gamma}{\eta} \left(1 - \frac{i\gamma}{\eta} \right) \\ &\quad \left. \left. \times \left(1 - \frac{\gamma}{\eta} \right)^{i-2} + \frac{\gamma}{\eta} q_{i-1}^{(20)}/q \right\} \left(\frac{\eta}{\eta+s} \right)^i \right\}. \end{aligned}$$

Combining the expressions of I and II, we get

$$\int_{0-}^{\infty} e^{-su} d\Phi_1(u) = -\frac{\sqrt{2}\rho v \langle f \phi' \rangle}{\sigma^2 \gamma} \left\{ \sum_{i=1}^{\infty} q_i^{(40)} \left(\frac{\eta}{\eta+s} \right)^i \right\}, \tag{B.15}$$

then the inversion of the Laplace transform implies Eq. (B.10). The proof is complete. □

Matching the coefficients in the above Proposition and Proposition 5.5, it is easy to find

$$q_i^* = q_i^{(41)} \quad \text{and} \quad q_i^{**} = q_i^{(40)}, \quad \forall i \ge 1.$$

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